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Soil moisture and nutrient sensor system

Abstract

A soil moisture and fertility sensor system is presented that includes an elongated probe having a plurality of sensor modules positioned along the length of the probe. Each sensor module includes a co-located sensors configured to take a moisture, temperature, and fertility measurements at varying depths of the soil. The probe is configured for wireless communication. The probe is configured to take moisture measurements, temperature measurements and/or nutrient measurements at different times so as to prevent interference between measurements. The probe includes a plurality of receptacles that receive the fertility sensor assembly cartridge that may be inserted into and removed from a receptacle so as to facilitate end-of-life replacement. In one arrangement, the fertility sensor assembly includes a reference sensor and a plurality of nutrient sensors that are each configured to sense the presence of a specific nutrient.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuing patent application of U.S. patent application Ser. No. 17/086,533 filed on Nov. 2, 2020 and titled SOIL MOISTURE AND NUTRIENT SENSOR SYSTEM, which claims priority to U.S. Provisional Application No. 62/930,182 filed on Nov. 4, 2019 and titled SOIL MOISTURE AND

NUTRIENT SENSOR SYSTEM, entirety of each of which is hereby incorporated by reference herein in its entirety, including any figures, tables, or drawings or other information.

FIELD OF THE DISCLOSURE

(1) This disclosure relates to agriculture. More specifically and without limitation, this disclosure relates to systems and methods of sensing moisture and nutrients in soil.

OVERVIEW OF THE DISCLOSURE

(2) Demands bestowed upon present day farmers are increasing by the day. Due to the cost of machinery, land, inputs, energy and labor, coupled with increasingly strict environmental demands, modern farmers must be vigilant to minimize inputs while maximizing yield while being as sustainable and environmentally friendly as possible.

(3) The success of any crop largely depends upon the crop having access to adequate moisture as well as adequate nutrients during the crop's life cycle. Despite the importance of moisture and nutrients, presently there is no convenient and cost effective way to comprehensively monitor moisture and nutrients across a farm field.

(4) Currently, farmers are forced to monitor in-season moisture and nutrients largely through a manual process. Many farmers simply drive around and/or walk through their fields to visually inspect the health of the crops. While this is somewhat effective, this method is labor intensive, time consuming, not-comprehensive and often leads to inconsistent and/or unscientific opinion-based results that are often not actionable in a timely manner.

(5) This visual inspection process may be supplemented by the collection of soil samples and/or tissue samples which are then sent to a lab to be scientifically analyzed. Soil sampling and tissue sampling provides more-scientific results than simply visually inspecting the crop. However, soil sampling and tissue sampling is extremely expensive, extremely time consuming and from a practical standpoint simply cannot be performed in a comprehensive manner across a large farm or field. As such, soil sampling and tissue sampling, at best, provides a farmer with a few data points from which the farmer must make drastic assumptions that are extrapolated across their entire farm or field.

(6) Both visual inspection and tissue sampling suffer from the same problem in that they are backward looking. That is, these processes require the effects of moisture levels and nutrient levels to appear in the crops, which is a trailing indicator. Therefore, this information only allows the farmer to be reactive instead of proactive. In addition, by the time the effects of moisture levels and nutrient levels appear in the crops often times irreparable damage and time loss has occurred that results in yield loss and suboptimal growth.

(7) To compensate for this lack of information or lack of timely information, many farmers are forced to over-apply various nutrients and/or moisture so to ensure the crops have access to adequate nutrients and/or moisture throughout the growing season. This over application results in excess cost of inputs. In addition, the over-application of nutrients results in nutrient runoff that results in high nitrate readings in downstream sources of drinking water as well as offsite contamination which has been linked to algae blooms in lakes and rivers and the Gulf of Mexico.

(8) As such, there is a need for an improved soil moisture and nutrient sensor system for agricultural use.

(9) Thus, it is a primary object of the disclosure to provide a soil moisture and nutrient sensor system that improves upon the state of the art.

(10) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is more environmentally friendly than existing systems and methods.

(11) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is more efficient than existing systems and methods.

(12) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is safer to use than existing systems and methods.

- (13) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides comprehensive moisture data across a field or farm.
- (14) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides comprehensive moisture data at varying depths.
- (15) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides comprehensive nutrient data across a field or farm.
- (16) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides comprehensive nutrient data at varying depths.
- (17) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides real-time data.
- (18) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides actionable data.
- (19) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides leading-indicator data.
- (20) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that allows a farmer to be proactive instead of reactive.
- (21) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that allows a farmer to maximize output.
- (22) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that allows a farmer to minimize inputs.
- (23) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that allows a farmer to optimize inputs and outputs.
- (24) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that gives a farmer unprecedented visibility to soil moisture and nutrients.
- (25) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that improves yields.
- (26) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that reduces costs.
- (27) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is relatively inexpensive.
- (28) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is simple to use.
- (29) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that can be used to sense nutrients for an entire growing season.
- (30) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that has replaceable nutrient sensors.
- (31) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is wireless.
- (32) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is battery powered.
- (33) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that helps a farmer meet increasingly strict environmental regulations.
- (34) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is easy to use.
- (35) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is easy to implement.
- (36) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that has a robust design.
- (37) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides charts and graphs of soil moisture and soil nutrients.

- (38) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides a farmer with increased confidence in their decisions.
- (39) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides accurate soil moisture readings.
- (40) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides accurate soil nutrient readings.
- (41) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that allows for the measurement of multiple nutrients.
- (42) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that normalizes moisture and nutrient readings with temperature.
- (43) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides co-located moisture, temperature and nutrient readings at various depths.
- (44) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that is cost effective to use.
- (45) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that can be used with any crop.
- (46) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that can be used with any nutrient.
- (47) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides highly repeatable results.
- (48) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that has a robust design.
- (49) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that provides highly accurate results.
- (50) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that has a long useful life.
- (51) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is easy to install.
- (52) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that has relatively few components.
- (53) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that has a minimum number of parts.
- (54) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that minimized nutrient runoff.
- (55) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that is environmentally friendly.
- (56) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that improves a farmer's efficiency.
- (57) Yet another object of the disclosure is to provide a soil moisture and nutrient sensor system that saves a farmer time.
- (58) Another object of the disclosure is to provide a soil moisture and nutrient sensor system that that provides scientific information regarding soil moisture and nutrients.
- (59) These and other objects, features, or advantages of the present disclosure will become apparent from the specification, claims and drawings.

SUMMARY OF THE DISCLOSURE

(60) A soil moisture and fertility sensor system is presented that includes an elongated probe having a plurality of sensor modules positioned along the length of the probe. Each sensor module includes a co-located moisture sensor assembly, temperature sensor assembly and fertility sensor assembly that take a moisture measurement, a temperature measurement and a fertility measurement at varying depths of the soil. The probe includes an on-board power source, which is

at least one battery, and a communication module that facilitates wireless communication. The probe is configured to take moisture measurements, temperature measurements and/or nutrient measurements at different times so as to prevent interference between measurements. The probe includes a plurality of receptacles that receive the fertility sensor assembly therein which is formed on a cartridge that may be inserted into and removed from a receptacle so as to facilitate end-of-life replacement. In one arrangement, the fertility sensor assembly includes a reference sensor and a plurality of nutrient sensors that are each configured to sense the presence of a specific nutrient. In this way, use of this system provides unprecedented soil moisture and nutrient information.

Description

BRIEF DESCRIPTION OF THE DRAWING

- (1) FIG. 1 shows a diagram of a soil moisture and fertility sensor system, in accordance with one or more embodiments.
- (2) FIG. 2 shows a front view of a wired probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments.
- (3) FIG. 3 shows a front view of a wireless probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments.
- (4) FIG. 4 shows a front view of a tube of a probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments; the view showing the tube with four sensor modules.
- (5) FIG. 5 shows a front view of a tube of a probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments; the view showing the tube with four sensor modules; the view showing a closeup of nutrition sensor assemblies of the sensor modules.
- (6) FIG. 6 shows tube segments of two probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments; the view showing the tube with six sensor modules; the view showing a closeup of nutrition sensor assemblies of the sensor modules.
- (7) FIG. 7 shows a front view of a of nutrition sensor assembly, in accordance with one or more embodiments; the view showing the nutrition sensor assembly having four nutrition sensors and one reference sensor.
- (8) FIG. 8 shows front, rear, and side views of a nutrition sensor assembly, in accordance with one or more embodiments.
- (9) FIG. 9 shows a side view of a of nutrition sensor assembly, in accordance with one or more embodiments.
- (10) FIG. 10 shows a soil cross section of a wired probe positioned in the ground for testing soil moisture and fertility sensor system, in accordance with one or more embodiments; the view showing sensitivity ranges of a plurality of sensor modules positioned in the probe.
- (11) FIG. 11 shows views of a gateway for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments; the left view showing a front exterior view of the gateway; the right view showing a front interior view of the gateway.
- (12) FIG. 12 shows a portion of a tube of a probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments.
- (13) FIG. 13 shows a front view of a wired probe for use in a soil moisture and fertility sensor system, in accordance with one or more embodiments; the view showing the tube of the wired probe with six sensor modules.
- (14) FIG. 14 shows a closeup view of a nutrition sensor assembly of a sensor module of the wired probe shown in FIG. 13, in accordance with one or more embodiments.
- (15) FIG. 15 shows an exploded lower rear left side perspective view of a nutrition sensor assembly of a sensor module of the wired probe shown in FIG. 13, in accordance with one or more

embodiments; the view showing a sensor cartridge positioned for insertion into a receptacle; the view showing a sensor PCB positioned for connection to the receptacle.

(16) FIG. 16 shows an exploded lower front right side perspective view of a nutrition sensor assembly of a sensor module of the wired probe shown in FIG. 13, in accordance with one or more embodiments; the view showing a sensor cartridge positioned for insertion into a receptacle; the view showing a sensor PCB positioned for connection to the receptacle.

DETAILED DESCRIPTION OF THE DISCLOSURE

(17) In the following detailed description of the embodiments, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific embodiments in which the disclosure may be practiced. The embodiments of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure. It will be understood by those skilled in the art that various changes in form and details may be made without departing from the principles and scope of the invention. It is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures. For instance, although aspects and features may be illustrated in or described with reference to certain figures or embodiments, it will be appreciated that features from one figure or embodiment may be combined with features of another figure or embodiment even though the combination is not explicitly shown or explicitly described as a combination. In the depicted embodiments, like reference numbers refer to like elements throughout the various drawings.

(18) It should be understood that any advantages and/or improvements discussed herein may not be provided by various disclosed embodiments, or implementations thereof. The contemplated embodiments are not so limited and should not be interpreted as being restricted to embodiments which provide such advantages or improvements. Similarly, it should be understood that various embodiments may not address all or any objects of the disclosure or objects of the invention that may be described herein. The contemplated embodiments are not so limited and should not be interpreted as being restricted to embodiments which address such objects of the disclosure or invention. Furthermore, although some disclosed embodiments may be described relative to specific materials, embodiments are not limited to the specific materials or apparatuses but only to their specific characteristics and capabilities and other materials and apparatuses can be substituted as is well understood by those skilled in the art in view of the present disclosure.

(19) It is to be understood that the terms such as “left, right, top, bottom, front, back, side, height, length, width, upper, lower, interior, exterior, inner, outer, and the like as may be used herein, merely describe points of reference and do not limit the present invention to any particular orientation or configuration.

(20) As used herein, the term “or” includes one or more of the associated listed items, such that “A or B” means “either A or B”. As used herein, the term “and” includes all combinations of one or more of the associated listed items, such that “A and B” means “A as well as B.” The use of “and/or” includes all combinations of one or more of the associated listed items, such that “A and/or B” includes “A but not B,” “B but not A,” and “A as well as B,” unless it is clearly indicated that only a single item, subgroup of items, or all items are present. The use of “etc.” is defined as “et cetera” and indicates the inclusion of all other elements belonging to the same group of the preceding items, in any “and/or” combination(s).

(21) As used herein, the singular forms “a,” “an,” and “the” are intended to include both the singular and plural forms, unless the language explicitly indicates otherwise. Indefinite articles like “a” and “an” introduce or refer to any modified term, both previously-introduced and not, while definite articles like “the” refer to a same previously-introduced term; as such, it is understood that

“a” or “an” modify items that are permitted to be previously-introduced or new, while definite articles modify an item that is the same as immediately previously presented. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including,” when used herein, specify the presence of stated features, characteristics, steps, operations, elements, and/or components, but do not themselves preclude the presence or addition of one or more other features, characteristics, steps, operations, elements, components, and/or groups thereof.

(22) It will be understood that when an element is referred to as being “connected,” “coupled,” “mated,” “attached,” “fixed,” etc. to another element, it can be directly connected to the other element, and/or intervening elements may be present. In contrast, when an element is referred to as being “directly connected,” “directly coupled,” “directly engaged” etc. to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” “engaged” versus “directly engaged,” etc.). Similarly, a term such as “operatively,” such as when used as “operatively connected” or “operatively engaged” is to be interpreted as connected or engaged, respectively, in any manner that facilitates operation, which may include being directly connected, indirectly connected, electronically connected, wirelessly connected or connected by any other manner, method or means that facilitates desired operation. Similarly, a term such as “communicatively connected” includes all variations of information exchange and routing between two electronic devices, including intermediary devices, networks, etc., connected wirelessly or not. Similarly, “connected” or other similar language particularly for electronic components is intended to mean connected by any means, either directly or indirectly, wired and/or wirelessly, such that electricity and/or information may be transmitted between the components.

(23) It will be understood that, although the ordinal terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited to any order by these terms unless specifically stated as such. These terms are used only to distinguish one element from another; where there are “second” or higher ordinals, there merely must be a number of elements, without necessarily any difference or other relationship. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of example embodiments or methods.

(24) Similarly, the structures and operations discussed herein may occur out of the order described and/or noted in the figures. For example, two operations and/or figures shown in succession may in fact be executed concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Similarly, individual operations within example methods described below may be executed repetitively, individually or sequentially, to provide looping or other series of operations aside from single operations described below. It should be presumed that any embodiment or method having features and functionality described below, in any workable combination, falls within the scope of example embodiments.

(25) As used herein, various disclosed embodiments may be primarily described in the context of soil measurements. However, the embodiments are not so limited. It is appreciated that the embodiments may be adapted for use in various other sensor applications, which may be improved by the disclosed structures, arrangements and/or methods. The system is merely shown and described as being used in the context of soil measurements for ease of description and as one of countless examples.

(26) System **10**:

(27) With reference to the figures, a soil moisture and fertility sensor system **10** (or simply system **10**), and method of use is presented. System **10** is formed of any suitable size, shape and design and is configured to facilitate the periodic measurement of moisture and nutrients in a farm field **12** at various locations around the field and at various depths.

(28) To be clear, while the term fertility sensor and the term soil fertility is used herein, these terms

are intended to also mean soil chemistry. That is, various constituents of soil that are important to the soil's health and ability to sustain crops may not necessarily be considered nutrients which may include sodium, pH, among others. As used herein the term fertility sensor and the term soil fertility as used herein is intended to cover all soil constituents including soil chemistry. As such, the soil moisture and fertility sensor system **10** presented herein is intended to sense soil moisture, soil fertility and soil chemistry.

(29) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** includes: a plurality of probes **14** having a tube **16** having an upper end **18** and a lower end **20**, a head **22**, a plurality of sensor modules **24** having a moisture sensor assembly **26**, a temperature sensor assembly **28** and fertility sensor assembly **30**, a gateway **32**, a cellular tower **34**, the internet **36**, the cloud **38**, a database **40**, a computing device **42**, displays **44** which are used to control inputs **46**, such as water **48** and nutrients **50** which are applied to the field **12**, among other components, pieces, systems and features, as is further described herein.

(30) Field **12**:

(31) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** is shown in use with one or more fields **12**. Field **12** is formed of any suitable size, shape and design and is configured to facilitate the growth of crops. In the arrangement shown, as one example, field **12** is a conventional farm field that consists of naturally deposited soil that is exposed to the elements such as that used to grow conventional crops such as corn, soybeans, wheat, hay, alfalfa, oats, cotton, vegetables, grapes, or any other crop. However, use of system **10** in association with a conventional farm or field **12** is only one of countless examples of use and is not intended to be limiting. Instead, it is hereby contemplated that system **10** is intended to be used not just with conventional farms and fields **12**, but also wherever crops or plants are grown. Additional examples include use with greenhouses, potted plants, orchards, vineyards, gardens, arboretums, grow houses, or any other place where plants or crops are grown.

(32) Probe **14**:

(33) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** includes the use of one or more probes **14**. Probe **14** is formed of any suitable size, shape and design and is configured to be inserted in the soil of the field **12**. Probe **14** is configured to sense moisture, temperature and nutrients in the soil of field **12** at various depths so as to provide a farmer with real-time actionable leading-indicator information so that the farmer may adjust the inputs **46**, such as water **48** and nutrients **50** before the lack of moisture and/or nutrients negatively affect plant growth. In the arrangement shown, as one example, probe **14** includes a tube **16** that extends a length from an upper end **18** to a lower end **20** and includes a head **22** connected to the upper end of tube **16**. However, any other configuration is hereby contemplated for use.

(34) Tube **16** (of Probe **14**):

(35) In the arrangement shown, as one example, probe **14** includes a tube **16**. Tube **16** is formed of any suitable size, shape and design and is configured to serve as the main housing of probe **14** and serves to house and hold and connect the other components of probe **14** while providing protection to the other components of probe **14**. Tube **16** is configured to be inserted within the soil of field **12** so as to facilitate the taking of moisture, temperature and nutrient measurements.

(36) In the arrangement shown, as one example, tube **16** extends a length from an upper end **18** to a lower end **20**. Tube **16** may extend any length. It is hereby contemplated that tube **16** may extend only a few inches to several feet, such as three feet, four feet, five feet, six feet, seven feet, eight feet, nine feet, ten feet, or more, or any range or length therein.

(37) In the arrangement shown, as one example, tube **16** is formed of a generally cylindrical tube-shaped member having a hollow interior, a pointed tip at its lower end **20** and head **22** connected at its upper end **18**.

(38) The pointed tip at lower end **20** helps to facilitate easy insertion of probe **14** into the soil of field **12**. The hollow interior is configured to receive, house and hold the other components of

probe **14** as is further described herein.

(39) While a cylindrical shaped tube **16** is shown, a cylindrical tube **16** is only one of countless possible shapes for tube **16**. Any other shaped member is hereby contemplated for use as tube **16** such as a square tube, a rectangular tube, an oval-shaped tube, a T-shaped member, an I-shaped member or any other shaped member.

(40) In the arrangement shown, as one example, tube **16** is formed of a single continuous member so as to eliminate seams that could allow for the infiltration of water and contaminants into the hollow interior of tube **16**. In an alternative arrangement, tube **16** is formed of a plurality of components that are connected together either in a removable manner, such as by screwing, bolting, clamping, snap-fitting or the like, or in a permanent manner such as by welding, gluing, or the like, or any combination thereof. Additionally or alternatively, in one or more arrangements tube **16** may be filled with a resin or another filler to restrict moisture or soil from entering the probe and damaging components.

(41) In one arrangement, the pointed tip at lower end **20** is formed as a continuous monolithic member with tube **16**. In an another arrangement, pointed tip at lower end **20** is assembled onto tube **16** either in a removable manner, such as by screwing, bolting, clamping, snap-fitting or the like, or in a permanent manner such as by welding, gluing, or the like, or any combination thereof.

(42) In the arrangement shown, as one example, tube **16** includes a plurality of sensor modules **24** that are positioned in spaced relation to one another along the length of tube **16**. In this way, when tube **16** is inserted vertically into the soil the spaced sensor modules **24** are positioned at different depths of the soil and the sensor modules **24** measure characteristics of the soil at different depths. In an arrangement shown, tube **16** of probe **14** is approximately twenty four inches long and includes sensor modules **24** at six positions along tube **16**. However, embodiments are not so limited. Rather, it is contemplated that in various arrangement, tube may be made to be any length and may include any number of sensor modules **24** at any number different positions along tube **16**.

(43) In the arrangement shown, as one example, tube **16** includes an upper-most sensor module **24** position a short distance below head **22**, a lower-most sensor module **24** positioned a distance above the tip at the lower end **20**, and a plurality of sensor modules **24** positioned at equal-spacing between the upper-most sensor module **24** and the lower-most sensor module **24**.

(44) In the arrangement shown, as one example, tube **16** includes a receptacle **52** positioned at the location of each sensor module **24**. In the arrangement shown, as one example, receptacle **52** is formed as an opening in the sidewall of tube **16** that is configured to receive a fertility sensor assembly **30** therein as is further described herein.

(45) In one arrangement, tube **16** is formed of a plastic material or non-metallic material that provides structural rigidity for protection purposes as well as being impermeable by water. In one arrangement a PVC (Polyvinyl Chloride) material is used, however any other non-metallic material is hereby contemplated for use. Alternatively, a metallic material may be used such as aluminum, iron, steel or any other metallic material or alloy.

(46) Head **22** (of Probe **14**):

(47) In the arrangement shown, as one example, probe **14** includes a head **22**. Head **22** is formed of any suitable size, shape and design and is configured to connect to the upper end **18** of tube **16** and enclose the upper end of probe **14**. In one arrangement, head **22** houses and/or holds and/or facilitates connection to all or a portion of various components of probe **14** such as lead **54**, controller assembly **56** including body **58**, processing circuit **60**, memory **62** having instructions **64**, such as software **66** and/or code **68**, a communication module **70** having antenna **72** and communication circuit **74**, an electrically connected power source **76**, among other components and parts and systems. When all or part of these or other components of probe **14** are not housed or held in head **22** they are housed or held within tube **16**, or alternatively these components may be housed or held within a telemetry unit **33** that is electrically connected to probe **14** through lead **54**.

(48) With reference to FIG. 2, a wired probe **14** is presented. In this arrangement, head **22**

facilitates connection to a lead **54**. Lead **54** is formed of any suitable size, shape and design and is configured to facilitate the wired transmission of information and/or power to and/or from probe **14** and gateway **32** or another component of the system **10**. Lead **54** may be a power cable, a coaxial cable, an Ethernet cable, an optical cable, a proprietary cable, or any other cable, lead, wire or the like or any combination thereof. Lead **54** facilitates the wired connection to another component of the system such as gateway **32** (or telemetry unit **33**), another probe **14**, an external power source such as a battery, a solar panel or line power, the internet, an intranet, or any other component of the system or combination thereof. In this arrangement, lead **54** extends into head **22** and electrically connects to some or all electronic components of probe **14** either directly or indirectly.

(49) With reference to FIG. **3**, a wireless probe **14** is presented. In this arrangement, head **22** is formed to house and hold all or a portion of communication module **70**. Communication module **70** is formed of any suitable size, shape and design and is configured to facilitate wireless communication of probe **14** with gateway **32**, cellular tower **34**, the internet **36**, the cloud **38**, the farmer's or user's computing device **42** or any other component of the system **10**.

(50) In the arrangement shown, as one example, communication module **70** is any device or combination of devices that facilitate wireless communication of probe **14**. In the arrangement shown, as one example, communication module **70** includes an antenna **72** and communication circuit **74** that facilitates wireless communication of probe **14**. In various arrangements, communication circuit **74** may be configured to communicate using any protocol such as Random Phase Multiple Access (RPMA), 802.11/Wi-Fi, Wi-Max, Bluetooth, Bluetooth Low Energy, UltraWideband (UWB), ZigBee, Zwave, GSM/EDGE, UMTS/HSPA+/HSDPA, CDMA, LTE, and/or FM/VHF/UHF networks or any other communication medium and/or protocol. A transmitter receiver is used for communication circuit **74** if one way communication is utilized in that a receiver is configured to transmit information only. In contrast, a transceiver is used for communication circuit **74** if two-way communication is utilized in that a transceiver sends as well as receives information.

(51) Various arrangements may utilize various different network topologies to facilitate connection between probes **14** and/or gateway **32** to facilitate communication or data from probes to database **40**. In the arrangement shown, as one example, a set of probes **14** are connected to a single gateway **32** in a star topology. However, embodiments are not so limited. Rather, it is contemplated that probes **14** and gateway(s) **32** may be connected in any network topology including but not limited to tree, mesh, star, ring, daisy chain, hybrid, and/or any other topology. In one or more arrangements, communication modules **70** of probes **14** and/or gateway are configured to form an adhoc network for communication of data.

(52) Communication module **70**, including communication circuit **74**, is connected to an antenna **72**. Antenna **72** is any device that facilitates the transmission and/or reception of the over-the-air signals, and may include a monopole antenna, dipole antenna, a loop antenna, a fractal antenna, or any other form of an antenna or combination thereof. Antenna **72** is configured to receive wireless signals from another component of the system **10**, such as gateway **32**, computing device **42** or another component, transmits these signals to the receiver/transceiver of communication module **70**. Communication module **70** processes these signals and then transmits these processed signals to processing circuit **60**, which processes these signals according to instructions stored in memory **62** as is further described herein.

(53) In one arrangement, head **22** is configured to sit just above the ground. That is, in this arrangement the lower end of head **22** is positioned at or just above the ground and the head **22** extends upward therefrom. This arrangement ensures wireless signals may be transmitted out of head **22** without interference of the ground. However, this arrangement exposes head **22** to the potential for damage, such as being run over by equipment, as head **22** protrudes out of the ground.

(54) In another arrangement, head **22** is configured to be inserted into the ground. That is, in this arrangement the upper end of head **22** is positioned at (or flush with) or just above or just below the

ground. This arrangement protects head 22 as head 22 is submerged into the ground and therefor if probe 14 is driven over, head 22 and probe 14 may not be damaged or may be somewhat protected from damage. However, this arrangement may present challenges as wireless signals may be inhibited from being transmitted out of head 22 due to interference of the ground. In addition, special attention may be needed to ensure that head 22 is properly sealed to prevent the infiltration of water and contaminants if head 22 is to be submerged within the soil.

(55) Controller Assembly 56, Processing Circuit 60, Memory 62, Instructions 64:

(56) In the arrangement shown, as one example, probe 14 includes a controller assembly 56. Controller assembly 56 is formed of any suitable size, shape and design and is configured to control operation of some or all of the electronic components of the probe 14. In the arrangement shown, as one example, controller assembly 56 includes one or more processing circuits 60, memory 62, or one or more memory devices, and instructions 64, which may be in the form of software 66 or code 68, among multiple other components and systems.

(57) In the arrangement shown, as one example, controller assembly 56 is electrically connected, either directly or indirectly, through wired connections, or wirelessly, to other electronic components positioned throughout probe 14 such as communication module 70, sensor modules 24, power source 76 and the like. In various arrangements, controller assembly 56 may be configured to communicate with the various other electronic components using any protocol such as Serial Data Interface 12 (SDI-12), UART, Serial Peripheral Interface, PCI/PCIE, Serial ATA, ARM Advanced Microcontroller Bus Architecture (AMBA), USB, Firewire, 802.11/Wi-Fi, Wi-Max, Bluetooth, Bluetooth Low Energy, Ultra Wideband (UWB), ZigBee, Zwave, GSM/EDGE, UMTS/HSPA+/HSDPA, CDMA, LTE, and/or FM/VHF/UHF networks or any other communication medium and/or protocol.

(58) In the arrangement shown, as one example, controller assembly 56 includes stored instructions 64, such as operational routines, that controller assembly 56 executes. Alternatively and/or in addition, in the arrangement shown, as one example, controller assembly 56 receives information and/or instructions 64 from gateway 32, a user-controlled computing device 42, a cellular tower 34, the internet 36, the cloud 38, or another component of the system 10. Controller assembly 56 processes this information according to instructions 64 stored in memory 62 and then outputs commands thereby controlling operation of the probe 14.

(59) In one or more arrangements, controller assembly 56 is configured to operate sensors modules 24 to take periodic measurements during operation. In one example arrangement, controller assembly 56 is configured to operate sensors modules 24 to take soil measurements every 15 minutes. However, embodiments are not so limited. Rather it is contemplated that various embodiments may be configured to take soil measurements at any interval or measurement schedule.

(60) In various different arrangements, controller assembly 56 may be configured to use communication circuit 74 to communicate soil measurements to gateway 32, cellular tower 34, the internet 36, the cloud 38, the farmer's or user's computing device 42 and or any other component of the system 10 at various intervals. In one or more arrangements, as one example, controller assembly 56 may be configured to use communication circuit 74 to communicate soil measurements in real time. Additionally or alternatively, in one or more arrangements, controller assembly 56 may be configured to store soil measurements in memory 62 and communicate multiple measurements to gateway 32, cellular tower 34, the internet 36, the cloud 38, the farmer's or user's computing device 42 and or any other component of the system 10 at a less frequent interval (e.g., hourly, daily, weekly, monthly, or any other interval). In this manner controller assembly 56 may save power by operating communication circuit 74 less frequently. Processing circuit 60 is formed of any suitable size, shape and design and is configured to receive and process information and output commands. In one or more arrangements, for example, such processing circuits includes a circuit specifically configured and arranged to carry out one or more of these or

related operations/activities. For example, processing circuit **60** may be discreet logic circuits or programmable logic circuits configured and arranged for implementing these operations/activities, as shown in the figures, and/or described herein. In certain embodiments, such a programmable circuit may include one or more programmable integrated circuits (e.g., field programmable gate arrays and/or programmable ICs). Additionally or alternatively, such a programmable circuit may include one or more processing circuits (e.g., a computer, microcontroller, system-on-chip, smart phone, server, and/or cloud computing resources). For instance, computer processing circuits may be programmed to execute a set (or sets) of instructions (and/or configuration data). The instructions (and/or configuration data) can be in the form of firmware or software stored in and accessible from a memory **62** (circuit). Certain embodiments are directed to a computer program product (e.g., nonvolatile memory device), which includes a machine or computer-readable medium having stored thereon instructions which may be executed by a computer (or other electronic device) to perform these operations/activities.

(61) Memory **62** is any form of information storage such as flash memory, ram memory, a hard drive, or any other form of memory or combination thereof. Memory **62** may be included as a part of or operably connected to processing circuit **60**. Controller assembly **56** may be a single component that is located at a single physical location. Alternatively, controller assembly **56** may be formed of multiple electronic components that are separated but communicatively connected to one another that act in concert with one another.

(62) Processing circuit **60** may be a single component that is located at a single physical location. Alternatively, processing circuit **60** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(63) Memory **62** may be a single component that is located at a single physical location. Alternatively, memory **62** may be formed of multiple electronic components that are separated but communicatively connected to one another that act in concert with one another.

(64) Processing circuit **60** and memory **62** may be a single joined component that is located at a single physical location, such as within head **22**, within tube **16**, or at any other location. Alternatively, processing circuit **60** and memory **62** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(65) In one arrangement, controller assembly **56**, which includes processing circuit **60**, memory **62** and instructions **64**, which may be in the form of software **66**, code **68** or any other form of information, controls operation of probe **14**. That is, in one arrangement, controller assembly **56** controls the operation of sensor modules **24** as well as the transmission of information through communication module **70**, as is further described herein.

(66) In one arrangement, wherein probe **14** is connected to telemetry unit **33** by lead **54**, controller assembly **56** and some or all of the related components described herein (processing circuit **60**, memory **62** and instructions **64**, communication module **70**, antenna **72**, communication circuit **74**, power source **76**, and the like) are contained within telemetry unit **33** which is configured to communicate with gateway **32**, the internet **36**, the cloud **38**, computing device **42** and/or another component of the system **10**.

(67) Sensor Modules **24**:

(68) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** includes a plurality of sensor modules **24**. Sensor modules **24** are formed of any suitable size, shape and design and are configured to facilitate the measurement of the moisture level, temperature, and/or nutrients in the soil surrounding probe **14**.

(69) In the arrangement shown, as one example, sensor modules **24** include a moisture sensor assembly **26**, a temperature sensor assembly **28** and a fertility sensor assembly **30**. In the arrangement shown, sensor modules **24** positioned along the length of tube **16** of probe **14** at equal spacing along the length of probe **14** such that when probe **14** is vertically inserted within the soil,

the sensor modules **24** are positioned such that sensor assemblies **26**, **28**, and/or **30** measures characteristics of the soil at different depths.

(70) In the arrangement shown, as one example, each of the sensor modules **24** include a co-located moisture sensor assembly **26**, a temperature sensor assembly **28**, and a fertility sensor assembly **30**. By co-locating the moisture sensor assembly **26**, the temperature sensor assembly **28** and the fertility sensor assembly **30** of each of the sensor modules **24** this ensures that accurate readings are being made of the soil moisture, temperature and fertility at the same or approximately the same position or vertical depth of the soil. In this way, the term co-locating is intended to mean located at approximately the same position on probe **14**.

(71) In one arrangement, the fertility sensor assembly **30** is positioned within or between the moisture sensor assembly **26** of each sensor module **24**.

(72) In the arrangement shown, as one example, every sensor module **24** includes a single co-located moisture sensor assembly **26**, temperature sensor assembly **28** and fertility sensor assembly **30**. However, it is hereby contemplated that each sensor module **24** may include any number of moisture sensor assemblies **26**, such as one, two, three, four, five, six, seven, eight, nine or ten or more. However, it is hereby contemplated that each sensor module **24** may include any number of temperature sensor assemblies **28**, such as one, two, three, four, five, six, seven, eight, nine or ten or more. However, it is hereby contemplated that each sensor module **24** may include any number of fertility sensor assemblies **30**, such as one, two, three, four, five, six, seven, eight, nine or ten or more. Whether sensor module **24** includes one or more moisture sensor assemblies **26**, temperature sensor assemblies **28** and/or fertility sensor assemblies **30**, these moisture sensor assemblies **26**, temperature sensor assemblies **28** and fertility sensor assemblies **30** are co-located within the sensor module **24**, which means they are located within a narrowly defined area or space where the moisture sensor assemblies **26**, temperature sensor assemblies **28** and fertility sensor assemblies **30** are located near one another so as to measure that soil at similar locations.

(73) Moisture Sensor Assembly **26** (of Sensor Module **24**):

(74) In the arrangement shown, as one example, sensor modules **24** of soil moisture and fertility sensor system **10** include a moisture sensor assembly **26**. Moisture sensor assembly **26** is formed of any suitable size, shape and design and is configured to measure the moisture content of the soil at the position of the moisture sensor assembly **26** which is co-located with the position of the temperature sensor assembly **28** and the fertility sensor assembly **30** of each sensor module **24**.

(75) Any form of a moistures sensor or moisture sensing system is hereby contemplated for use as moisture sensor assembly **26**. In addition, while only one moisture sensor assembly **26** is shown in use with each sensor module **24**, any number of moisture sensor assemblies **26** are hereby contemplated for use with each sensor module **24**.

(76) In the arrangement shown, as one example, moisture sensor assembly **26** is formed of what is known as a capacitance sensor. In electrical engineering, capacitive sensing (sometimes capacitance sensing) is a technology, based on capacitive coupling, that can detect and measure anything that is conductive or has a dielectric different from air. In the arrangement shown, as one example, when moisture sensor assembly **26** is a capacitive moisture sensor, moisture sensor assembly **26** includes a positive plate **78**, a negative plate **80** and a space or dielectric **82** positioned between the positive plate **78** and the negative plate **80**.

(77) A capacitive moisture sensor works by measuring the changes in capacitance caused by the changes in the dielectric **82**. A capacitive moisture sensor does not measure moisture directly (pure water does not conduct electricity well), instead it measures the ions that are dissolved in the moisture. These ions and their concentration can be affected by a number of factors, for example adding fertilizer for instance will decrease the resistance of the soil. Capacitive measuring basically measures the dielectric that is formed by the soil and the water is the most important factor that affects the dielectric.

(78) Noncontact capacitive sensors work by measuring changes in an electrical property called

capacitance. Capacitance describes how two conductive objects (positive plate **78** and negative plate **80**) with a space between them (space or dielectric **82**) respond to a voltage difference applied to the positive plate **78** and negative plate **80**. When a voltage is applied to the positive plate **78** and negative plate **80**, an electric field is created in the space or dielectric **82** between them causing positive and negative charges to collect on each of the positive plate **78** and negative plate **80**.

(79) In one arrangement, capacitive sensors use an alternating voltage which causes the charges to continually reverse their positions. The moving of the charges creates an alternating electric current which is detected by a sensor. The amount of current flow is determined by the capacitance, and the capacitance is determined by the area and proximity of the conductive objects. Larger and closer objects cause greater current than smaller and more distant objects. The capacitance is also affected by the type of nonconductive material in the space or dielectric **82** between the positive plate **78** and negative plate **80**.

(80) In the arrangement shown, capacitive-based moisture sensor assemblies **26** at different heights along tube **16** of probe **14** to measure moisture content at different soil depths. It has been surprisingly discovered that placement of capacitive-based moisture sensor assemblies **26** at positioned are separated by at least four inches avoids interference between capacitive-based moisture sensor assemblies **26** and improves sensitivity. However, it is contemplated that in one or more arrangements capacitive-based moisture sensor assemblies **26** may be separated by any distance along tube **16** of probe **14**.

(81) Capacitive measuring has some advantages. Namely, capacitive measuring avoids corrosion of the positive plate **78** and the negative plate **80**. Capacitive measuring also gives a better reading of the moisture content of the soil as opposed to using a resistive soil moisture sensor. Capacitive measuring avoids corrosion of the positive plate **78** and the negative plate **80** since the positive plate **78** and the negative plate **80** of the capacitor are not exposed to the soil. As such, there is no corrosion of the positive plate **78** and the negative plate **80**.

(82) In the arrangement shown, as one example, positive plate **78** and negative plate **80** are formed of electrical conductors. In the arrangement shown, as one example, positive plate **78** and negative plate **80** are formed of electrical plates. In the arrangement shown, as one example, positive plate **78** and negative plate **80** are positioned within the hollow interior of tube **16** and have an exterior surface that is in engagement with or in close proximity to the interior surface of tube **16**.

(83) In the arrangement shown, as one example, positive plate **78** and negative plate **80** extend in a continuous and uninterrupted manner around the entire circumference of the interior surface of tube **16**. That is, in this arrangement, positive plate **78** and negative plate **80** extend three hundred and sixty degrees around the probe **14**, and/or around the interior surface of the hollow interior of tube **16**.

(84) In one arrangement, positive plate **78** and negative plate **80** are formed on and/or into an insert **84** that is configured to be inserted within the hollow interior of tube **16**. In the arrangement shown, as one example, insert **84** extends a length between an upper end **86** and a lower end **88** and is sized and shaped to fit within the hollow interior of tube **16** with close and tight tolerances. In the arrangement shown, as one example, insert **84** has an exterior size and shape that closely matches the interior size and shape of tube **16**. That is, when tube **16** is cylindrical in shape, insert **84** is cylindrical in shape; when tube **16** is square in shape, insert **84** is square in shape, and so on.

(85) As the moisture level or humidity of the soil between the positive plate **78** and negative plate **80** changes, the measured capacitance changes. These changes in capacitance of the soil in the space or dielectric **82** between positive plate **78** and negative plate **80** are extrapolated using an algorithm to determine the true moisture content of the soil at the location of that particular sensor module **24**. In one arrangement, the temperature measurement taken by the co-located temperature sensor assembly **28** is used in association with the capacitance measurement taken by the co-located moisture sensor assembly **26** to normalize the moisture measurement for that particular sensor module **24**. This helps to take reduce or eliminate the affect that temperature has on

capacitance readings.

(86) In the arrangement shown, as one example, the temperature sensor assembly **28** and fertility sensor assembly **30** of each sensor module **24** are positioned between the positive plate **78** and negative plate **80** of the moisture sensor assembly **26**. Or, said another way, in the arrangement shown, as one example, the temperature sensor assembly **28** and the fertility sensor assembly **30** of each sensor module **24** are positioned within the space or dielectric **82** of the moisture sensor assembly **26**.

(87) In an alternative arrangement, temperature sensor assembly **28** and fertility sensor assembly **30** of each sensor module **24** are positioned adjacent the positive plate **78** or negative plate **80** of the moisture sensor assembly **26**. In an alternative arrangement, a temperature sensor assembly **28** is positioned on the outside of each of the positive plate **78** and negative plate **80**. In an alternative arrangement, a fertility sensor assembly **30** is positioned on the outside of each of the positive plate **78** and negative plate **80**.

(88) Notably, while a single positive plate **78** and a single negative plate **80** is shown in use, any number of positive plates **78** and negative plates **80** are hereby contemplated for use. As one example, each moisture sensor assembly **26** may include a single positive plate **78** and two negative plates **80**, one on each side of the positive plate **78**. As another example, each moisture sensor assembly **26** may include a single negative plate **80** and two positive plates **78**, one on each side of the negative plate **80**. As another example, each moisture sensor assembly **26** may include a two positive plates **78** and two negative plates **80** which are adjacent each other. Any other number of positive plates **78** and negative plates **80** are hereby contemplated for use as is any other arrangement of these positive plates **78** and negative plates **80**.

(89) In one arrangement, moisture sensor assembly **26** uses Frequency Domain Reflectometry (FDR) or alternatively Time-Domain Reflectometry (TDR). These technologies do not measure actual soil moisture levels but soil tension, electro-conductivity, di-electric constant or resistance.

(90) Temperature Sensor Assembly **28** (of Sensor Module **24**):

(91) In the arrangement shown, as one example, sensor modules **24** of soil moisture and fertility sensor system **10** include a temperature sensor assembly **28**. Temperature sensor assembly **28** is formed of any suitable size, shape and design and is configured to measure the temperature of the soil at the position of the temperature sensor assembly **28** which is co-located with the position of the moisture sensor assembly **26** and the fertility sensor assembly **30** of each sensor module **24**.

(92) Any form of a temperature sensor or temperature sensing system is hereby contemplated for use as temperature sensor assembly **28**. In addition, while only one temperature sensor assembly **28** is shown in use with each sensor module **24**, any number of temperature sensor assemblies **28** are hereby contemplated for use with each sensor module **24**.

(93) In the arrangement shown, as one example, temperature sensor assembly **28** is formed of what is known as a thermometer, or a digital thermometer or an electronic thermometer. However, any other form of a temperature sensor is hereby contemplated for use including thermocouples, resistance temperature detectors (RTDs), thermistors, semiconductor based integrated circuits (ICs), and/or any other form of a temperature sensor.

(94) Thermocouples are made by joining two dissimilar metal wires together. This causes a Seebeck Effect. The Seebeck Effect is a phenomenon in which a temperature difference of two dissimilar conductors produces a voltage difference between the two substances. It is this voltage difference that can be measured and used to calculate the temperature.

(95) RTDs operate under the principle that as temperature changes, the resistance of any metal changes as well. This difference in resistance is what RTD temperature sensors are based on. An RTD is a resistor with well-defined resistance vs. temperature characteristics. Platinum is the most common and accurate material used to make RTDs.

(96) Thermistors are similar to RTDs in that temperature changes cause measurable resistance changes. Thermistors are usually made from a polymer or ceramic material. In most cases,

thermistors are cheaper but are also less accurate than RTDs. Most thermistors are available in two wire configurations. The NTC (Negative Temperature Coefficient) thermistor is the most commonly used thermistor for temperature measurement application. An NTC thermistor's resistance decreases as the temperature increases. Thermistors have a non-linear temperature resistance relationship. This requires a significant correction to interpret the data correctly.

(97) Semiconductor based temperature sensor ICs come in two different types: local temperature sensor and remote digital temperature sensor. Local temperature sensors are ICs that measure their own die temperature by using the physical properties of a transistor. Remote digital temperature sensors measure the temperature of an external transistor.

(98) In the arrangement shown, as one example, like positive plate **78** and negative plate **80**, temperature sensor assembly **28** is formed as part of insert **84** and is positioned within the hollow interior of tube **16**. In this way, tube **16** protects temperature sensor assembly **28**. In one arrangement, as one example, temperature sensor assembly **28** includes a probe that extends outward from or is flush with the exterior surface of tube **16** so as to ensure direct contact with the soil to ensure accurate measurements.

(99) In the arrangement shown, as one example, like fertility sensor assembly **30**, temperature sensor assembly **28** is positioned within moisture sensor assembly **26**. That is, in one arrangement, temperature sensor assembly is positioned between the positive plate **78** and negative plate **80** of moisture sensor assembly **26**.

(100) The temperature measurements provided by temperature sensor assemblies **28** are in and of themselves valuable information. In addition, in one arrangement, temperature measurements provided by temperature sensor assembly **28** are used to normalize the readings from moisture sensor assembly **26** and/or fertility sensor assembly **30**. That is, the readings from temperature sensor assembly **28** and moisture sensor assembly **26** are used as inputs into an algorithm or program performed by processing circuit **60** of controller assembly **56** or another processing device or processing circuit of another component of the system **10** which normalizes the reading from moisture sensor assembly **26** to provide an accurate measurement of the moisture level in the soil. Similarly, the readings from temperature sensor assembly **28** and fertility sensor assembly **30** are used as inputs into an algorithm or program performed by processing circuit **60** of controller assembly **56** or another processing device or processing circuit of another component of the system **10** which normalizes the reading from fertility sensor assembly **30** to provide an accurate measurement of the nutrient level in the soil.

(101) Fertility Sensor Assembly **30** (of Sensor Module **24**):

(102) In the arrangement shown, as one example, sensor modules **24** of soil moisture and fertility sensor system **10** include a fertility sensor assembly **30**. Fertility sensor assembly **30** is formed of any suitable size, shape and design and is configured to measure one or more characteristics of the soil, such as various nutrient levels, ion levels, pH and/or other characteristics that relate to fertility and soil health and productivity.

(103) Any form of a fertility sensor or fertility sensing system is hereby contemplated for use as fertility sensor assembly **30**. In addition, while only one fertility sensor assembly **30** is shown in use with each sensor module **24**, any number of fertility sensor assemblies **30** are hereby contemplated for use with each sensor module **24**.

(104) Cartridge **90**:

(105) In the arrangement shown, as one example, fertility sensor assembly **30** includes a cartridge **90**. Cartridge **90** is formed of any suitable size, shape and design and is configured to fit within receptacle **52** of probe **14**. Cartridge **90** is configured to be easily installed into receptacle **52** of probe **14** prior to use. Cartridge **90** is configured to be held within receptacle **52** of probe **14** during use. Cartridge **90** is configured to be easily removed from receptacle **52** after use and after cartridge **90** has exceeded its useful lifespan. That is, cartridge **90** is configured to be easily inserted within and removed from receptacle **52** so as to facilitate a removable and disposable fertility sensor

assembly **30**.

(106) In the arrangement shown, as one example, cartridge **90** of fertility sensor assembly **30** includes a body **92**. Body **92** of cartridge **90** of fertility sensor assembly **30** is formed of any suitable size, shape and design and is configured to be the main structural element of cartridge **90** and is configured to house and hold and connect the other components of cartridge **90** of fertility sensor assembly **30** while providing structural support and rigidity.

(107) In the arrangement shown, as one example, body **92** is formed of a generally planar member having a generally flat exterior surface **94** and a generally flat interior surface **96** and an exterior peripheral edge **98**. In the arrangement shown, as one example, exterior surface **94** and interior surface **96** extend in approximate parallel planar spaced relation to one another. In the arrangement shown, as one example, exterior peripheral edge **98** extends in an approximate perpendicular manner to the planes formed by exterior surface **94** and interior surface **96**. In the arrangement shown, as one example, body **92** is generally rectangular shaped when viewed from above or below and includes generally straight sidewalls **100** that extend in approximate parallel spaced relation to one another and includes generally straight end walls **102** that extend in approximate parallel spaced relation to one another. In the arrangement shown, as one example, sidewalls **100** and end walls **102** extend in approximate perpendicular relation to one another thereby forming a generally square or rectangular shaped member.

(108) In the arrangement shown, as one example, sidewalls **100** and end walls **102** connect to one another at rounded corners. Collectively, sidewalls **100**, end walls **102**, and the rounded corners positioned there between, form peripheral edge **98** of body **92** of cartridge **90** of fertility sensor assembly **30**.

(109) In the arrangement shown, as one example, body **92** is formed of a printed circuit board (PCB), a substrate, or another form of a device that provides structural rigidity as well as selective electrical connections and selective electrical isolations.

(110) Sealing Member **104**:

(111) In the arrangement shown, as one example, cartridge **90** includes a sealing member **104**. Sealing member **104** is formed of any suitable size, shape and design and is configured to allow for the insertion of cartridge **90** into receptacle **52**, as well as facilitate removal of cartridge **90** from receptacle **52**, as well as facilitate forming a seal between cartridge **90** of fertility sensor assembly **30** and receptacle **52** of probe **14**.

(112) In the arrangement shown, as one example, sealing member **104** is a flexible, compressible or otherwise malleable member that forms a seal between cartridge **90** and receptacle **52**. In the arrangement shown, as one example, sealing member **104** is an O-ring that extends around the exterior peripheral edge **98** of body **92** of cartridge **90** in a continuous and uninterrupted manner. In one arrangement, body **92** includes a wall that extends rearward from interior surface **96** a distance. This wall may be formed as part of sealing member **104**. Alternatively, this wall may be a separate component from sealing member **104**. This wall provides alignment and additional surface area of engagement between the walls of receptacle **52** and the exterior peripheral edge **98** of cartridge **90** thereby providing increased sealing capabilities.

(113) Electrical Contacts **106**:

(114) In the arrangement shown, as one example, cartridge **90** includes one or more electrical contacts **106** positioned in its interior surface **96** that electrically connect to one or more sensors in its exterior surface **94** through traces or leads that extend through body **92** of cartridge **90**. These sensors include reference sensor **108** and nutrient sensors **110**.

(115) Electrical contacts **106** are formed of any suitable size, shape and design and are configured to facilitate the establishment of an electrical connection between cartridge **90** and the other components of probe **14**. In the arrangement shown, as one example, electrical contacts **106** of cartridge **90** are sized and shaped to engage electrical contacts **112** in receptacle **52** of probe **14**. In one arrangement, electrical contacts **106** of cartridge **90** are metallic and/or conductive pads that

receive and/or engage electrical contacts **112** of receptacle **52** which may be formed of compressible electrical contacts, such as electrically conductive foams or “pogo pins” which are spring loaded metallic pins that facilitate the establishment of an electrical connection between two components. In an alternative arrangement, electrical contacts **106** of cartridge **90** may be formed of compressible electrical contacts, such as what are known as “pogo pins” which are spring loaded metallic pins that facilitate the establishment of an electrical connection between two components and electrical contacts **112** of receptacle **52** are metallic and/or conductive pads. The use of electrical contact pads on one of cartridge **90** and receptacle **52** and the use of compressible electrical contacts on the other of cartridge **90** and receptacle **52** facilitates the easy installation and removal of cartridge **90** into receptacle **52** while facilitating a secure and durable electrical connection between cartridge **90** and receptacle **52**.

(116) In one arrangement, when electrical contacts **106** of cartridge **90** are spring loaded pogo pins, when cartridge **90** is replaced, the pogo pins are replaced. This is beneficial as pogo pins do, overtime, wear out and break, this may be especially true when cartridge **90**, and the attached pogo pins are exposed to moisture, dirt and other contaminants. And as such, by replacing cartridge **90** the pogo pins are replaced. This may provide durability, robustness, repeatability, and a long useful life.

(117) Reference Sensor **108** and Nutrient Sensors **110**:

(118) In the arrangement shown, as one example, cartridge **90** includes at least one reference sensor **108** and at least one nutrient sensor **110** positioned in its exterior surface **94** that are each electrically connected to an electrical contact **106** in the interior surface **96** of cartridge **90**.

Reference sensor **108** and nutrient sensors **110** are formed of any suitable size, shape and design and are configured to sense or detect the concentration of various nutrients in the soil. In the arrangement shown, as one example, reference sensor **108** and nutrient sensors **110** are what are known as ion selective electrodes. An ion-selective electrode (ISE), also known as a specific ion electrode (SIE), is a transducer (or sensor) that converts the activity of a specific ion dissolved in a solution into an electrical potential. The voltage is theoretically dependent on the logarithm of the ionic activity, according to the Nernst equation.

(119) In the arrangement shown, as one example, each cartridge **90** includes one reference sensor **108** and four nutrient sensors **110**, however any number of reference sensors **108** and nutrient sensors **110** are hereby contemplated for use.

(120) In the arrangement shown, as one example, reference sensor **108** and nutrient sensors **110** are formed of gold pads that are connected to the exterior surface **94** of cartridge **90**. Gold pads, whether solid gold or plated gold, are useful in that gold does not oxidize. However, any other conductive material is hereby contemplated for use.

(121) In the arrangement shown, as one example, reference sensor **108** is formed an elongated rectangular pad that extends along one side of cartridge **90**. In the arrangement shown, as one example, reference sensor **108** extends the majority of the length of the end wall **102** to end wall **102** length of cartridge **90**. However, any other length or shape or configuration is hereby contemplated for use.

(122) In the arrangement shown, as one example, nutrient sensors **110** are formed of generally cylindrical shaped pads that are aligned along one side of cartridge **90**, opposite reference sensor **108**, in a row. In the arrangement shown, as one example, four nutrient sensors **110** are shown, adjacent reference sensor **108** and as such, nutrient sensors **110** are approximately one quarter the size of the reference sensor **108**. However, any other length or shape or configuration is hereby contemplated for use.

(123) In the arrangement shown, as one example, reference sensor **108** and nutrient sensors **110** include a polyimide layer **114** and a graphene layer **116**.

(124) Polyimide Layer **114** and Graphene Layer **116** and Perforations **118**:

(125) In the arrangement shown, as one example, reference sensor **108** and nutrient sensors **110**

include a polyimide layer **114** and a graphene layer **116**.

(126) In the arrangement shown, as one example, polyimide layer **114** covers the entire exterior surface **94** of cartridge **90**. In an alternative arrangement, polyimide layer **114** covers a portion of the exterior surface **94** of cartridge **90**. In an alternative arrangement, polyimide layer **114** covers the entire surface of reference sensor **108** and nutrient sensors **110**.

(127) Polyimide (sometimes abbreviated PI) is a polymer of imide monomers. With their high heat-resistance, polyimides enjoy diverse applications in roles demanding rugged organic materials, e.g. high temperature fuel cells, displays, and various military roles. One example polyimide is Kapton, which is produced by condensation of pyromellitic dianhydride and 4,4'-oxydianiline.

(128) In the arrangement shown, as one example, polyimide layer **114** serves as an electrical insulator to the conductive reference sensor **108** and nutrient sensors **110**.

(129) In the arrangement shown, as one example, polyimide layer **114** is formed of a Kapton tape. Kapton is a polyimide film developed by DuPont in the late 1960s that remains stable across a wide range of temperatures, from -269 to $+400^{\circ}\text{C}$. (-452 to 752°F ; 4 to 673 K). Kapton is used in, among other things, flexible printed circuits (flexible electronics) and thermal blankets used on spacecraft, satellites, and various space instruments.

(130) The chemical name for Kapton K and HN is poly (4,4'-oxydiphenylene-pyromellitimide). It is produced from the condensation of pyromellitic dianhydride and 4,4'-oxydiphenylamine. Kapton synthesis is an example of the use of a dianhydride in step polymerization. The intermediate polymer, known as a "poly(amic acid)", is soluble because of strong hydrogen bonds to the polar solvents usually employed in the reaction. The ring closure is carried out at high temperatures (200 - 300°C . (392 - 572°F ; 473 - 573 K)).

(131) In the arrangement shown, as one example, polyimide layer **114** is applied to the exterior surface **94** of cartridge **90** as a Kapton tape. In this arrangement, an adhesive, or an adhesive layer is positioned between the polyimide layer **114** and the exterior surface **94** of cartridge **90** to ensure adhesion of the polyimide layer **114** to the exterior surface **94** of cartridge **90**.

(132) Any other form of material is hereby contemplated for use as polyimide layer **114**, particularly any other form of material that can be converted into graphene.

(133) In some arrangements, as one example, a plurality of perforations **118** or micro-perforations are formed in polyimide layer **114** prior to application of polyimide layer **114** onto the exterior surface **94** of cartridge **90**. Perforations **118** are formed of any suitable size, shape and design and are formed of openings that extend through the polyimide layer **114** from its upper surface to its lower surface.

(134) In the arrangement shown, as one example, perforations **118** are generally small circular openings in polyimide layer **114**, however any other size, shape and design is hereby contemplated for use. In the arrangement shown, as one example, perforations **118** are placed in polyimide layer **114** above reference sensor **108** and nutrient sensors **110**. In this way, when polyimide layer **114** is applied to the exterior surface **94** of cartridge **90**, perforations **118** land on top of reference sensor **108** and nutrient sensors **110**. In this way, the openings formed by perforations **118** provide access to the exterior surface of reference sensor **108** and nutrient sensors **110**.

(135) After polyimide layer **114** is applied to the exterior surface **94** of cartridge **90**, sections **120** are treated with a laser. As the laser interacts with the exterior surface of polyimide layer **114** a layer of graphene, or graphene layer **116** is formed. In one arrangement a CO₂ laser is used to form graphene layer **116** from polyimide layer **114**. However, any other form of a laser is hereby contemplated for use.

(136) In one or more arrangements, copper traces are laid around the edge of the polyimide layer **114**. The Lasing of the polyimide electronically welds the graphene (LIG) to the trace. These traces extend through body **92** of cartridge **90** to connect with electrical contacts **106** of cartridge **90**, in which case perforations **118** may be omitted. The LIG and with copper trace creates a tighter hydrophobic seal, which may provide more accurate and more stable readings.

(137) Forming graphene layer **116** by applying a laser to polyimide layer **114** is called Laser-induced Graphene (LIG). Laser-induced graphene (LIG) is a 3D porous material prepared by direct laser writing with a CO₂ or other laser on carbon materials in ambient atmosphere. This technique combines 3D graphene preparation and patterning into a single step without the need for wet chemical steps.

(138) Graphene, the atomically-thin honeycomb carbon lattice, is a highly conducting 2D material whose exposed electronic structure offers a platform for chemical and biological sensing. Its biocompatible, flexible and chemically inert nature associated with the lack of dangling bonds, offers opportunities for direct interfacing with biological molecules. Combined with its exceptional electronic and optical properties, this promotes graphene as a unique platform for bioelectronics.

(139) In one or more arrangements, LIG is produced by applying a laser over the carbon material in a nitrogen gas atmosphere. However, embodiments are not so limited. Rather it is contemplated that various arrangements may use other LIG formation techniques. For example, it has been surprisingly discovered two or more applications of a DPI laser at different power levels and/or or at different angles in a normal environment (no Nitrogen gas) generates a more highly conductive and robust sensor.

(140) In one arrangement, the laser is applied in sections **120** of polyimide layer **114**. That is, wherever the laser is applied, the upper surface or the exterior surface of polyimide layer **114** is converted into a layer of graphene or graphene layer **116**. However, this graphene layer **116** is extremely thin, and does not extend all the way through polyimide layer **114**. As such, despite the fact that graphene layer **116** is conductive, because of the underlying polyimide layer **114** the reference sensor **108** and nutrient sensors **110** positioned below the graphene layer **116** and polyimide layer **114** would remain insulated if it were not for perforations **118** that extend through polyimide layer **114**.

(141) In the arrangement shown, as one example, when reference sensor **108** is generally rectangular in shape, the section **120** covers the interior portion of reference sensor **108**. That is, in this arrangement, the section **120** of reference sensor **108** is a slightly smaller rectangular shape that is positioned just inward a distance from the exterior peripheral edge of reference sensor **108**. Similarly, in the arrangement shown, as one example, when nutrient sensors **110** are generally circular in shape, the section **120** covers the interior portion of nutrient sensors **110**. That is, in this example arrangement, the section **120** of nutrient sensors **110** is a slightly smaller circular shape that is positioned just inward a distance from the exterior peripheral edge of nutrient sensors **110**.

(142) In the arrangement shown, as one example, when a laser is applied to sections **120** of polyimide layer **114**, the upper surface or exterior surface of polyimide layer **114** is converted into graphene layer **116**. This graphene layer **116** is conductive. In the arrangement shown, as one example, when a laser is applied to the edges of perforations **118** a graphene layer **116** is formed along the edges of perforations **118**.

(143) This graphene layer **116** along the edges of perforations **118** forms a conduit or connector or conductor from the graphene layer **116** in the exterior surface of polyimide layer **114** down to the reference sensor **108** or nutrient sensor **110** positioned below the polyimide layer **114**. In addition to serving as an electrical connection to reference sensor **108** or nutrient sensor **110**, the intersection at the lower end of perforations **118** is fused to the reference sensor **108** or nutrient sensor **110** when the laser is applied to polyimide layer **114** thereby forming a graphene layer **116**. That is, when the laser hits the intersection of the lower end of the polyimide layer **114** of a perforation, a portion of the polyimide layer **114** and a portion of the material of the reference sensor **108** or nutrient sensor **110** are melted and/or mixed, and/or fused and/or welded together as the graphene layer **116** is formed thereby securely affixing the graphene layer **116**, as well as the polyimide layer **114**, to the underlying reference sensor **108** or nutrient sensor **110**. This prevents delamination of the graphene layer **116**, as well as the polyimide layer **114**, from the underlying reference sensor **108** or nutrient sensor **110**. This prevents electrical disconnection of the graphene layer **116** from

the underlying reference sensor **108** or nutrient sensor **110**. For lack of a better term, by providing perforations **118** in polyimide layer **114** and applying a laser to the polyimide layer **114** and forming a graphene layer **116** that fuses to the underlying reference sensor **108** or nutrient sensor **110** this plugs the graphene layer **116** into the underlying reference sensor **108** or nutrient sensor **110**.

(144) In the arrangement shown, as one example, each reference sensor **108** and nutrient sensor **110** include a plurality of perforations **118** within each section **120** of graphene layer **116** and as such, graphene layer **116** is electrically and physically connected to the underlying reference sensor **108** or nutrient sensor **110** at each perforation **118**. In this way a graphene based electrode is formed or said another way, a graphene based sensor is formed. When ions come in direct contact with the graphene layer **116**, the ions change the potential of the electrode by charge transfer, which is directly proportional to the ion concentration. These graphene based electrodes or graphene based sensors are substantially less expensive to manufacture in terms of fabrication as compared to comparable alternatives, and they are amenable to miniaturization, low-energy consumption, as well as diverse applications.

(145) In the arrangement shown, reference sensor **108** is an open sensor in that it does not have an ion selective membrane **122** covering the reference sensor **108**. As such, all ions are able to reach the reference sensor **108**. This provides a constant reference potential, and potentiostat, which records the open circuit potential (OCP) which is used to normalize the readings from the nutrient sensors **110** which include an ion selective membrane **122**.

(146) As an alternative to forming graphene along the sides of a perforation **118** to form an electrical contact with the underlying reference sensor **108** or nutrient sensor **110**, in one arrangement perforations **118** are filled with a conductive material. This conductive material may be plated, deposited, injected or inserted by any other manner. In one arrangement a gold paste or silver paste is used. This conductive material completes the connection between the exposed surface of the underlying reference sensor **108** or nutrient sensor **110** within the perforation **118** and the graphene layer **116** of sections **120** on top of the polyimide layer **114**. Any other manner, method or means of connecting the graphene layer **116** with the underlying reference sensor **108** or nutrient sensor **110** is hereby contemplated for use.

(147) Ion Selective Membrane **122**:

(148) In the arrangement shown, as one example, an ion selective membrane **122** is applied to nutrient sensors **110** but not reference sensors **108**. Ion selective membrane **122** is formed of any suitable size, shape and design and is configured to prevent all but select nutrients or ions to pass through the ion selective membrane **122**. In this way, ion selective membrane **122** is what is known as a selectively permeable membrane in that it allows some materials, ions, nutrients, chemistries, molecules, constituents, etc. to pass through the ion selective membrane **122** while preventing other materials, ions, nutrients, chemistries, molecules, constituents, etc. from passing through the ion selective membrane **122**.

(149) In one or more arrangements, ion selective membrane **122** is formed of a PVC-based ion selective membrane that is selective to Ammonium, Calcium, Chlorine, Hydrogen, Nitrate, Magnesium, Phosphorus, Potassium, or Sodium ions or the like or any other materials, ions, nutrients, chemistries, molecules, constituents, or elements or range thereof or combination thereof.

(150) In one or more arrangements, ion selective membrane **122** may be formed of PVC layer having a porous scaffold configuration on a molecular level. This may also be referred to as a nano-porous polymer membrane. In one or more arrangements, ion selective membrane **122** is what is known as an ionophore-doped membranes which may be used for functionalisation in ion sensing technologies such as ion-selective electrodes (ISEs). Doping the ion selective membrane **122** in this manner allows only selected ions through the porous ion selective membrane **122**.

(151) In the arrangement shown, as one example, where multiple nutrient sensors **110** are present on a cartridge **90**, all of the nutrient sensors **110** may be configured to detect the same ion, which

provides redundancy which may lead to higher accuracy and confidence in the reading. Alternatively, where multiple nutrient sensors **110** are present on a cartridge **90**, each of the nutrient sensors **110** may be configured to detect a different ion, which provides the ability to sense the concentration of multiple nutrients using only a single cartridge **90**. Another benefit of the cartridge and the co-located multiple nutrient sensors **110** is that the nutrient sensors **110** are able to detect various ion levels or nutrient levels at approximately the same position in the soil. This may be repeated by providing multiple cartridges **90** along the length of probe **14**.

(152) As one example, a single cartridge **90** having a reference sensor **108** and four nutrient sensors **110** may have an ion selective membrane **122** covering the first nutrient sensor **110** that is limited to hydrogen, an ion selective membrane **122** covering the second nutrient sensor **110** that is limited to Ammonium or Nitrate, an ion selective membrane **122** covering the third nutrient sensor **110** that is limited to Phosphorus, and an ion selective membrane **122** covering the first nutrient sensor **110** that is limited to Potassium, and in this way, a single cartridge **90** may be used to detect the pH, as well as the N, P, K concentrations of the soil.

(153) In this example arrangement, the nutrient sensors **110** collect ion level readings that pass through the ion specific membranes **122** which prohibit other ions from passing through the ion specific membrane **122**. These ion readings, nutrient readings, and/or fertility readings are then temperature compensated using the temperature reading from the temperature sensor assembly **28** and the raw data is transmitted in a data string with both Scaled Frequency Units (SFU) and temperature readings and moisture levels. In this example arrangement, this data is then stored and processed to equate parts per million (PPM) to pounds per acre (PPA) by moisture and temperature compensation along with normalization using information from reference sensor **108**.

(154) In the arrangement shown, as one example, a ceramic or Borosilicate glass membrane **124** is applied over reference sensor **108** and nutrient sensors **110**.

(155) In one arrangement, reference sensors **108** includes an ion selective membrane **122** similar to that identified above with respect to nutrient sensors **110**. However, the ion selective membrane **122** applied to reference sensor **108** includes a silver/silver chloride ink or paste, which is very stable. This provides a reference reading or reference voltage or reference potential that the readings of the nutrient sensors **110** may be compared to.

(156) Membrane **124**:

(157) In the arrangement shown, as one example, a membrane **124** is applied over reference sensor **108** and nutrient sensors **110**. Membrane **124** is formed of any suitable size, shape and design and is configured to cover reference sensor **108** and nutrient sensors **110** so as to prevent soil particles, organic matter or other contaminants from adhering to the reference sensor **108** and nutrient sensors **110**. In one arrangement, membrane **124** is formed of a ceramic or Borosilicate glass, however any other form of a material or a membrane is hereby contemplated for use that allows moisture, nutrients and the like to pass through the membrane **124** while preventing soil particles, organic matter or other contaminants from adhering to the reference sensor **108** and nutrient sensors **110**.

(158) Once cartridge **90** is formed, cartridge **90** is configured to be inserted within a receptacle **52** of probe **14**.

(159) Receptacle **52**:

(160) In the arrangement shown, as one example, probe **14** includes one or more receptacles **52**. Receptacles **52** are formed of any suitable size, shape and design and are configured to receive a cartridge **90** therein with close and tight tolerances while facilitating an electrical connection of cartridge **90** with sensor PCB **176** and/or controller assembly **56**, while forming a seal with cartridge **90** and while allowing the exterior surface **94** of cartridge **90** to engage the soil, among other features, objectives and advantages. Receptacles **52** are configured to hold a cartridge **90** therein during use. Receptacle **52** is configured to allow for the easy installation of cartridge **90** into receptacle. Receptacle is configured to allow for the easy removal of cartridge **90** from receptacle

52 after use and after cartridge **90** has exceeded its useful lifespan. This removable cartridge **90** feature allows the probe **14** to be reused by simply replacing the cartridges **90**. This provides added emphasis on ensuring that the cartridges **90** are as low-cost as possible so as to ensure disposable cartridges **90** is economically feasible, which is something the system **10** and configuration presented herein accomplishes.

(161) To facilitate this easy installation and removal of a cartridge **90**, receptacle **52** and cartridge **90** are formed of complementary sizes and shapes. That is, in the arrangement shown, as one example, when cartridge **90** is generally rectangular in shape, receptacle **52** is similarly rectangular in shape and is sized and shaped to facilitate the easy mating insertion of cartridge **90** within receptacle **52**. Any other shape is hereby contemplated for use for receptacle **52** as well as cartridge **90**.

(162) In the arrangement shown, as one example, receptacle **52** includes a sidewall **126** that extends inward from the exterior surface of probe **14** and connects to a rear wall **128**. Rear wall **128** of receptacle **52** includes electrical contacts **112** therein that connect to the electrical contacts **106** in the interior surface **96** of cartridge **90**. In the arrangement shown, electrical contacts **112** are electrically connected to electrical contacts **113** on an interior side of receptacle **52**. In the arrangement shown, as one example, electrical contacts **113** connect with an edge connector **178** of sensor PCB **176** to facilitate communication of sensor measurements to controller assembly **56**. Sensor PCB **176** is formed of any suitable size shape and design and is configured to communicate with controller assembly **56** and perform soil measurements with sensors **108** and **110**.

Alternatively, in some arrangements, electrical contacts **113** may be connected directly to controller assembly and sensor PCB **176** is omitted.

(163) In the arrangement shown, electrical contacts **112** of receptacle **52** are compressible pogo pins compressible then contact electrical contacts **106** of cartridge **90** when cartridge **90** is inserted within receptacle **52**. However, the opposite arrangement is hereby contemplated for use wherein the receptacle **52** includes the electrical contact pads that connect with pogo pins **106** of cartridge

(164) In the arrangement shown, as one example, when cartridge **90** is inserted within a receptacle **52**, sealing member **104** engages and seals to the sidewall **126** of receptacle **52** thereby preventing moisture and contaminants from entering receptacle **52**.

(165) In one or more arrangements, a locking member **130** is associated with receptacle **52**. Locking member **130** is formed of any suitable size, shape and design and is configured to lock cartridge **90** in receptacle **52** once installed. In one or more arrangements, locking member **130** may be formed of a screw or bolt or other fastener that extends through or engages cartridge **90** locks or tightens it to probe **14**. In one or more arrangements, locking member **130** may be formed of a snap fit feature, a detent, a friction fit feature, a bracket, a pin, or any other component or system that secures or locks or helps to hold cartridge **90** into receptacle **52**. This locking member **130** may also serve to ensure a good and strong seal between cartridge **90** and receptacle **52** so as to prevent moisture and contaminant infiltration.

(166) In the arrangement shown, as one example, to help further prevent moisture and contaminant infiltration, a potting compound **174** is placed within receptacle **52**. In one or more arrangements, potting compound **174** may be formed of any material that prevents or reduces the infiltration of water into receptacle **52** and/or behind cartridge **90**. In one or more arrangements, potting compound **174** is a silicone, dielectric grease, glycerin grease, or any other compound that prevents or reduces the infiltration and effects of moisture.

(167) Power Source **76**:

(168) In the arrangement shown, as one example, probe **14** includes a power source **76**. Power source **76** is formed of any suitable size, shape and design and is configured to connect to and provide power for the electronic components of the probe **14**. In one arrangement, power source **76** is formed of one or more batteries that are positioned inside tube **16** and/or head **22**. In an alternative arrangement, power source **76** is positioned within an external telemetry unit **33** or

another component that is connected to probe **14** by lead **54**.

(169) One of the benefits of having power source **76** positioned within the probe **14** is this makes probe **14** fully wireless. In this arrangement, probe **14** is inserted within the soil without any external components.

(170) In an alternative arrangement, probe **14** may be connected by lead **54** to an external component such as gateway **32**, telemetry unit **33** or the like.

(171) Gateway **32** and Telemetry Unit **33**:

(172) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** includes one or more gateways **32** and/or telemetry units **33**. In one arrangement, gateway **32** and telemetry unit **33** are similar components with one difference being telemetry unit **33** is configured to connect directly to a probe **14** by way of lead **54** and in this way serve as an above-ground communication and control unit for probe **14**. Note that a telemetry unit **33** is only used with a probe **14** when probe **14** does not have the components of telemetry unit **33** on board the probe **14** already. In contrast, gateway **32** is configured to serve as a base station that sends and receives wireless signals (or in some cases wired signals) between one or more probes **14** and/or telemetry units **33** installed throughout a farm or field. Notably, in one arrangement both a gateway **32** and telemetry unit **33** may serve as a repeater in a mesh network established between a plurality of probes **14** of the system **10**, and as such, each probe **14**, telemetry unit **33** and/or gateway **32** may serve as a repeater or essentially another gateway **32**.

(173) For purposes of avoiding redundancy both gateway **32** and telemetry unit **33** are both considered to have the following components and operate in a similar manner, unless specifically identified otherwise.

(174) In the arrangement shown, as one example, gateways **32** and/or telemetry units **33** include a housing **132**. Housing **132** is formed of any suitable size, shape and design and is configured and is configured to house and hold the components of gateways **32** and/or telemetry units **33** as is described herein. In the arrangement shown, as one example, housing **132** is formed of an enclosure having a main body **134** and a cover **136** that opens and closes and seals a hollow interior **138**. In the arrangement shown, as one example, hollow interior **138** houses controller assembly **140**.

(175) Controller assembly **140** is formed of any suitable size, shape and design and is configured to control operation of some or all of the electronic components of gateway **32** and/or telemetry unit **33** and/or probe **14** if a probe **14** or multiple probes **14** are attached. In the arrangement shown, as one example, controller assembly **140** includes one or more processing circuits **142**, memory **144**, or one or more memory devices, and instructions **146**, which may be in the form of software **148** or code **150**, among multiple other components and systems.

(176) In the arrangement shown, as one example, controller assembly **140** is electrically connected, either directly or indirectly, through wired connections, or wirelessly, to other electronic components of the system **10** including other gateways **32** and/or telemetry units **33** and probes **14**. In the arrangement shown, as one example, controller assembly **140** includes stored instructions **146**, such as operational routines, that controller assembly **140** executes. Alternatively and/or in addition, in the arrangement shown, as one example, controller assembly **140** receives information and/or instructions **146** from other gateways **32** and/or telemetry units **33**, a user-controlled computing device **42**, a cellular tower **34**, the internet **36**, the cloud **38**, or another component of the system **10**. Controller assembly **140** processes this information according to instructions **146** stored in memory **144** and then outputs commands thereby controlling operation of the controller assembly **140**, any attached probes **14** or other components of the system **10**.

(177) In one or more arrangements, processing circuit **142** may be any computing device that receives and processes information and outputs commands according to instructions stored in memory **144**. Memory **144** is any form of information storage such as flash memory, ram memory, a hard drive, or any other form of memory or combination thereof. Memory **144** may be included

as a part of or operably connected to processing circuit **142**. Controller assembly **140** may be a single component that is located at a single physical location. Alternatively, controller assembly **140** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(178) Processing circuit **142** may be a single component that is located at a single physical location. Alternatively, processing circuit **142** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(179) Memory **144** may be a single component that is located at a single physical location. Alternatively, memory **144** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(180) Processing circuit **142** and memory **144** may be a single joined component that is located at a single physical location, such as within housing **132**, or at any other location. Alternatively, processing circuit **142** and memory **144** may be formed of multiple electronic components that are separated but electrically connected to one another that act in concert with one another.

(181) In one arrangement, controller assembly **140**, which includes processing circuit **142**, memory **144** and instructions **146**, which may be in the form of software **148**, code **150** or any other form of information, controls operation of and/or works in concert with any connected probes **14** and/or other gateways **32** and/or telemetry units **33**. That is, in one arrangement, controller assembly **140** controls the operation of connected probes **14** and/or other gateways **32** and/or telemetry units **33** as well as the transmission of information through communication module **152**, as is further described herein.

(182) Communication module **152** is formed of any suitable size, shape and design and is configured to facilitate wireless communication of gateway **32** and/or telemetry unit **33** with other gateways **32** and/or telemetry units **33**, cellular tower **34**, the internet **36**, the cloud **38**, the farmer's or user's computing device **42** or any other component of the system **10**.

(183) In the arrangement shown, as one example, communication module **152** is any device or combination of devices that facilitate wireless communication of gateway **32** and/or telemetry unit **33**. In the arrangement shown, as one example, communication module **152** includes an antenna **154** and radio **156** that facilitates wireless communication of gateway **32** and/or telemetry unit **33** and/or an attached probe **14** or other component of the system **10**.

(184) A receiver is used for radio **156** if one way communication is utilized in that a receiver is configured to transmit information only. In contrast, a transceiver is used for radio **156** if two-way communication is utilized in that a transceiver sends as well as receives information.

(185) Communication module **152**, including radio **156**, is connected to an antenna **154**. Antenna **154** is any device that facilitates the transmission and/or reception of the over-the-air signals, and may include a monopole antenna, dipole antenna, a loop antenna, a fractal antenna, or any other form of an antenna or combination thereof. Antenna **154** is configured to receive wireless signals from another component of the system **10**, such as another gateway **32** and/or telemetry unit **33**, computing device **42** or another component, transmits these signals to the receiver/transceiver of communication module **152**, which processes these signals and then transmits these processed signals to processing circuit **142** which processes these signals according to instructions **146** stored in memory **144** as is further described herein.

(186) Communication and Calculations:

(187) In the arrangement shown, as one example, a plurality of probes **14** are installed across a field or farm. Each of the probes **14** periodically and/or according to a plan, prescription, schedule or other program or a user initiated request, makes a moisture measurement using moisture sensor assemblies **26**, a temperature measurement using temperature sensor assemblies **28** and a fertility measurement using fertility sensor assemblies **30** at the location of each sensor modules **24**. Note, however, that not all sensor modules **24** include a fertility sensor assembly **30**. In contrast, with reference to FIG. **6**, insert **84** only includes a fertility sensor assembly **30** as part of the top three

sensor modules **24**. This is because fertility and/or nutrients are primarily concentrated in the upper regions of the soil. Or, farmers are only interested in knowing the fertility and/or nutrient concentrations in the upper regions of the soil.

(188) In the arrangement shown, as one example, each of the plurality of probes **14** include an on-board communication module **70** or are connected by lead **54** to a telemetry unit **33** or gateway **32**. In one arrangement, as one example, probe **14** makes a moisture measurement using moisture sensor assemblies **26**, a temperature measurement using temperature sensor assemblies **28** and a fertility measurement using fertility sensor assemblies **30** at the location of each sensor modules **24** (collectively a measurement). These measurements are transmitted by wireless communication through communication module **70** of probe **14** or communication module **152** of telemetry unit **33**. In one arrangement, a plurality of measurements are stored in memory **62/144** and transmitted in bulk so as to save power and reduce the number of data transmissions. This process of collecting readings, which are saved for future communications, is commonly referred to as EDA (Environmental Data Acquisition). This data is then converted to a communications protocol that is used to transmit the data wirelessly when desired.

(189) These wireless signals are then received by a generally centrally positioned gateway **32** which then re-transmits them to cellular tower **34**, or another communication installation or system. Cellular tower **34** then transmits the information through one or more information networks **158**, which may be a wireless network, a wired network, or any combination thereof, to the internet **36** and/or the cloud **38**. In one or more arrangements, these measurements are stored in database **40**.

(190) One of the benefits of system **10** is that the co-location of moisture sensor assemblies **26**, temperature sensor assemblies **28** and fertility sensor assemblies **30** at the location of each sensor modules **24** allows for normalization of readings from moisture sensor assemblies **26**, a fertility sensor assemblies **30** at the location of each sensor modules **24** as temperature has an effect on moisture level readings from moisture sensor assemblies **26** and moisture and temperature have effect on readings from fertility sensor assemblies **30**. In addition, readings from reference sensor **108** are used to normalize readings from ion specific nutrient sensors **110** having ion selective membranes **122** thereon. Using an algorithm or other program, this normalization may occur at the on-board controller assembly **56**, or connected controller assembly **140** of a connected gateway **32** or telemetry unit **33**, or alternatively the raw data may be transmitted to database **40** and the normalization or calculations may occur in association with database **40**.

(191) To facilitate this data processing, in the arrangement shown, as one example, database **40** includes or is connected to or is associated with a processing circuit **160**, memory **162**, or one or more memory devices, and instructions **164**, which may be in the form of software or code or algorithms, among other components and systems. Alternatively and/or in addition to processing circuit **160**, memory **162**, and/or instructions **164** being associated with database **40**, processing circuit **160**, memory **162**, and/or instructions **164** may be associated with computing device **42**, or alternatively processing circuit **160**, memory **162**, and/or instructions **164** may be associated with both database and computing device **42** which operate in a cooperative manner with one another.

(192) To avoid redundancy, processing circuit **160**, memory **162**, and instructions **164**, whether associated with database **40** or computing device **42**, are similar to that described herein with respect to probe **14**, gateway **32** and/or telemetry unit **33**. In this arrangement, processing circuit **160** in association with memory **162** and instructions **164** performs calculations to convert the raw readings from probes **14** to normalized readings which may be in the form of parts per million (PPM) or pounds per acre (PPA) or any other format that is useful and valuable to the farmer.

(193) In one or more arrangements, processing circuit **160**, memory **162**, and instructions **164** of database **40** and/or computing device **42** may be configured to additionally or alternatively perform various data analytics processes on data stored in database **40** to derive additional data metrics pertinent to assessment of soil measurements. In some arrangements, the data analytics processes may correlate data in database **40** with data in third party data sources to derive such data metrics.

As an illustrative example, in one or more arrangements the data analytics processes may cross correlate data in database with publicly available data sets (e.g., provided by various government agencies), which monitor temperature, ground water, and/or weather, for example, to derive additional data metrics. In one or more embodiments, the analytics processes may be configured to analyze the data and learn, over time, identify trends or various conditions of concern. Additionally or alternatively, analytics processes may be configured to analyze the data and learn over time predictive models, for example, to predict future soil conditions that will result from a proposed soil treatment based on based on current measurements. Such learning by the analytics processes may include, for example, generation and refinement of classifiers and/or state machines configured to map input data values to such trends or conditions of concern. In various embodiments, analysis by the analytics processes may include various guided and/or unguided artificial intelligence and/or machine learning techniques including, but not limited to: neural networks, genetic algorithms, support vector machines, k-means, kernel regression, discriminant analysis and/or various combinations thereof. In different implementations, analysis may be performed locally, remotely, or a combination thereof.

(194) In association with and/or in addition to performing the normalization operation converting raw data to normalized and useful data and/or analytics processes, in one arrangement database **40** and/or computing device **42** communicatively connected to database **40** generates displays **44** of the measurement information from probes **14**. These displays **44** may be any form of presentation of the information or measurements generated by probes **14**. This may include charts, graphs, diagrams, tables, spreadsheets, or the like. These displays **44** may be integrated with or overlaid on or otherwise associated with other forms of information such as yield maps, application maps, seeding maps, treatment maps, spraying maps, or any other form of information that provides greater information and insight to the farmer. These displays **44** may be interactive and may be manipulated by the farmer or a crop-insight manager or agronomist or the like in any way to yield practically any form of insight into field health.

(195) In the arrangement shown, as one example, these displays **44** are displayed on the interactive display **168** of computing device **42** which may take the form of a desktop or laptop computer, a smart phone, a tablet or any other form of a computing device. Computing device **42** is then used by the farmer or a crop-insight manager or agronomist or the like to provide actionable insights.

(196) In response to the insights provided by the information provided by probes **14**, analytics processes, and/or displays **166**, the farmer then may take action by amending the inputs **46**. If the land is irrigated, the farmer may amend the amount of water **48** applied to the field. Alternatively, the farmer may amend the nutrients **50** applied to the field.

(197) One of the benefits of the system **10** is that probes **14** may provide multiple nutrient-specific measurements at multiple depths through the use of fertility sensor assemblies **30** having multiple nutrient sensors **110** each having an ion selective membrane **122** is that the farmer may specifically address nutrient deficiencies. That is, if the field is low in potassium, the farmer may apply only potassium; if the field is low in nitrogen, the farmer may apply only nitrogen; and so on. As such, for the first time, the farmer may in real-time have access to moisture and specific nutrient information across a farm or field. With nutrient specific information the farmer may specifically address deficiencies. This reduces cost by eliminating the use of unneeded nutrients. This also improves the environment as this reduces the potential for nutrient runoff and offsite contamination.

(198) Simultaneous Operation-Individual Operation:

(199) In the arrangement shown, as one example, probes **14** include multiple sensor modules **24** each having a moisture sensor assembly **26** and a temperature sensor assembly **28** and some of them having a fertility sensor assembly **30** which may include a reference sensor **108** as well as multiple nutrient sensors **110**. In one arrangement, when a measurement is taken, all of the measurements are taken simultaneously. That is, all of the sensors of sensor modules **24** are

operated at the same time. That is each of the moisture sensor assemblies **26** and the temperature sensor assemblies **28** and the reference sensor **108** as well as multiple nutrient sensors **110** of the fertility sensor assemblies **30** are energized and take a measurement at the same time or approximately at the same time. This arrangement may provide some benefits such speed and being able to perform a complete measurement within a minimum amount of time. Another benefit may be minimal battery usage.

(200) However, performing multiple measurements simultaneously provides some risks and drawbacks. Namely, the energization of multiple sensors located close to one another may cause interference that skews or obfuscates the readings.

(201) In an alternative arrangement, multiple sensors are operated at different times. In one arrangement, all of the moisture sensor assemblies **26** are operated at the same time while the temperature sensor assemblies **28** and fertility sensor assemblies **30** are not energized. This arrangement makes the moisture measurement at all sensor modules **24** at the same time.

(202) In one arrangement, all of the temperature sensor assemblies **28** are operated at the same time while the moisture sensor assemblies **26** and fertility sensor assemblies **30** are not energized. This arrangement makes the temperature measurement at all sensor modules **24** at the same time.

(203) In one arrangement, all of the fertility sensor assemblies **30** are operated at the same time while the moisture sensor assemblies **26** and temperature sensor assemblies **28** are not energized. This arrangement makes all the fertility measurements at all sensor modules **24** at the same time.

(204) As an alternative to this arrangement, as fertility sensor assemblies **30** include a reference sensor **108** and multiple nutrient sensors **110**, in one arrangement, only one of the reference sensor **108** and multiple nutrient sensors **110** of each fertility sensor assembly **30** is operated at the same time while the others of the reference sensor **108** and multiple nutrient sensors **110** of each fertility sensor assembly **30** are not energized. This way, at any one time, only one of the reference sensors **108** and multiple nutrient sensors **110** of each fertility sensor assembly **30** is energized.

(205) In an alternative arrangement, only one of the multiple sensor modules **24** are energized at any one time. In one arrangement, as one example, if a probe **14** includes three sensor modules **24**, each having a moisture sensor assembly **26**, a temperature sensor assembly **28** and a fertility sensor assembly **30** having a reference sensor **108** and multiple nutrient sensors **110**, the measurement is performed in the following manner: All of the sensors of the first sensor module **24** are energized simultaneously; All of the sensors of the second sensor module **24** are energized simultaneously; All of the sensors of the third sensor module **24** are energized simultaneously.

(206) In this way, all of the measurements are performed in a manner wherein the operation of one sensor module **24** does not affect the other sensor modules **24**. However, the simultaneous energization of the sensors within each sensor module **24** may affect the readings of the other sensors within each sensor module **24**.

(207) In another arrangement, as one example, if a probe **14** includes three sensor modules **24**, each having a moisture sensor assembly **26**, a temperature sensor assembly **28**, and a fertility sensor assembly **30** having a reference sensor **108** and multiple nutrient sensors **110**, only one of the moisture sensor assembly **26**, temperature sensor assembly **28**, and fertility sensor assembly **30** of one of the sensor modules **24** is energized and/or read at any one time. As an example, the measurement is performed in the following manner: The first sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized and read; The temperature sensor assembly **28** is energized and read; The reference sensor **108** and multiple nutrient sensors **110** of the fertility sensor assembly **30** are read simultaneously or nearly simultaneously; The second sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized and read; The temperature sensor assembly **28** is energized and read; The reference sensor **108** and multiple nutrient sensors **110** of the fertility sensor assembly **30** are read simultaneously or nearly simultaneously; and The third sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized; The temperature sensor assembly

28 is energized; and The reference sensor **108** and multiple nutrient sensors **110** of the fertility sensor assembly **30** are energized simultaneously.

(208) In this way, all of the measurements are performed in a manner wherein the operation of one sensor module **24** does not affect the other sensor modules **24**. In addition, the operation of the moisture sensor assembly **26**, temperature sensor assembly **28** and fertility sensor assembly **30** does not affect the other of the moisture sensor assembly **26**, temperature sensor assembly **28** and fertility sensor assembly **30** within each sensor module **24**. However, the simultaneous energization of the reference sensor **108** and multiple nutrient sensors **110** of each fertility sensor assembly **30** may affect the readings of the other sensors within the fertility sensor assembly **30**.

(209) In another arrangement, as one example, if a probe **14** includes three sensor modules **24**, each having a moisture sensor assembly **26**, a temperature sensor assembly **28** and a fertility sensor assembly **30** having a reference sensor **108** and multiple nutrient sensors **110** only one of the moisture sensor assembly **26**, temperature sensor assembly **28** and fertility sensor assembly **30** of one of the sensor modules **24** is energized at any one time with each of the reference sensor **108** and multiple nutrient sensors **110** of the fertility sensor assembly **30** energized individually. As an example, the measurement is performed in the following manner: The first sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized and read; The temperature sensor assembly **28** is energized and read; The reference sensor **108** of fertility sensor assembly **30** is read; Each of the multiple nutrient sensors **110** of fertility sensor assembly **30** are individually read sequentially; The second sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized and read; The temperature sensor assembly **28** is energized and read; The reference sensor **108** of fertility sensor assembly **30** is read; Each of the multiple nutrient sensors **110** of fertility sensor assembly **30** are individually read sequentially; and The third sensor module **24** is measured in the following manner: The moisture sensor assembly **26** is energized and read; The temperature sensor assembly **28** is energized and read; The reference sensor **108** of fertility sensor assembly **30** is read; and Each of the multiple nutrient sensors **110** of fertility sensor assembly **30** are individually read sequentially.

(210) In this way, all of the measurements are performed in a manner wherein the operation of any sensor of any sensor module **24** does not affect any other measurement of any other sensor modules **24**. This is perhaps the most cautious manner of operating the probe **14**. However, this manner of operation takes the longest and may increase power draw as functional components of the probe **14** are operating for a longer period of time as opposed to going into a sleep mode between periodic measurements.

(211) Any other manner of operation of probe **14** is hereby contemplated for use.

(212) Assembly and Installation:

(213) In the arrangement shown, as one example, soil moisture and fertility sensor system **10** is assembled and installed in the following manner: Cartridges **90** having the desired ion selective membranes **122** are installed into each of the receptacles **52** along the length of tube **16** of probe **14**. This installation is accomplished by inserting a cartridge **90** into each receptacle **52** by aligning the peripheral edge **98** of cartridge **90** with the walls of receptacle **52** and pressing the cartridge **90** into the receptacle **52** until the electrical contacts **106** of cartridge **90** engage and electrically connect to the electrical contacts **112** of receptacle **52**. In this position, the exterior surface of cartridge **90** or more specifically the exterior surface of reference sensor **108** and/or nutrient sensors **110** are in approximate flush alignment with the exterior surface of tube **16** of probe **14**.

(214) Alternatively, it is also contemplated that, in one or more arrangements, the exterior surface of cartridges **90** may be slightly recessed to or slightly proud of the exterior surface of tube **16**. However, cartridges **90** are configured to operate through direct connection with the soil. As such, while recessing the exterior surface of cartridges **90** provides some protection to scratching and damaging cartridge **90** upon insertion of the probe into the soil, this recessing poses problems when it comes to ensuring a direct connection between the soil and the exterior surface of the cartridges

90. In contrast, having the exterior surface of cartridges **90** protrude slightly from the exterior surface of the tube **16** helps to ensure a direct connection with the soil. However, having the exterior surface of the cartridges **90** protrude from the exterior surface of the tube **16** exposes the thin and delicate membranes on the surface of the reference sensor **108** and nutrient sensors **110** to damage during insertion of the probe **14** into the soil.

(215) In one arrangement, to facilitate installation of probe **14** into the soil of a field, first the location is selected. Second a hole is formed by suitable measures in the soil of the field. In one arrangement, care is taken when forming the hole that the sides of the hole have as crisp an edge as possible so as to ensure the best possible direct engagement between the soil and the cartridges **90** of probe **14**. As examples, this hole may be formed by cutting, plugging, drilling or any other manner, method or means.

(216) Once the hole is formed, the probe **14** is carefully inserted within the hole until the fully inserted. Once probe **14** is inserted into the soil, in one arrangement, as one example, to help ensure direct contact between cartridges **90** the soil, a slurry is formed in the soil around the probe **14** and/or the soil is emulsified around the probe **14** thereby ensuring the soil makes direct contact with the cartridges **90**. In some arrangements, slurry may be formed by mixing a specified ratio of soil with water. This will ensure a consistent slurry is used for all sensors.

(217) However, injecting a slurry or emulsifying the soil affects the moisture content of the soil around the probe **14** which affects the moisture readings for some time after installation until the soil around the probe **14** re-establishes equilibrium with the surrounding soil. Typically, such equilibrium occurs within a few days.

(218) Once probe **14** is inserted, the sensor modules **24** measure the characteristics of the soil in areas **170** around the sensor modules **24**. Each sensor module **24** provides information regarding the characteristics of the soil at different depths, including moisture level, temperature, and the concentration of nutrients.

(219) After the useful life of the cartridges **90** has been exceeded, which in one arrangement is the length of a growing season which may be as long as nine-months, probe **14** is removed from the soil and the cartridges **90** are removed from each receptacle **52**. New cartridges **90** are inserted into each receptacle **52** and the probe **14** is ready to be used again.

(220) Biasing Member **172**:

(221) As mentioned herein, reference sensor **108** and nutrient sensors **110** are formed of what are known as direct contact ion sensors which operate through the direct contact of the soil with the sensors. As such, it is important to ensure that direct contact between the soil and reference sensor **108** and nutrient sensors **110** is established. In one arrangement, to ensure this direct contact occurs, probe **14** includes a biasing member **172**. Biasing member **172** is formed of any suitable size, shape and design and is configured to selectively force cartridges **90** outward from tube **16** after tube **16** is inserted within the soil.

(222) In this arrangement, cartridges **90** may be recessed to the exterior surface of tube **16** during installation of the tube **16** into the soil. Then, after installation, biasing member **172** is engaged or operated which forces cartridges **90** outward until the exterior surface of cartridges **90** engage the soil in flat and flush and constant and repeatable and reproducible engagement. This insures accurate and consistent readings between sensor modules **24** and between probes **14**. This also eases the installation process and eliminates the need to inject a slurry around the probe **14** or emulsify the soil around the probe **14**.

(223) In one arrangement, biasing member **172** is a spring loaded system that once engaged applies a spring loaded force thereby forcing cartridges **90** outward until they engage the soil. In another arrangement, biasing member **172** is a mechanical cam system that when engaged includes an arm that forces cartridges **90** outward until they engage the soil. In another arrangement, biasing member **172** includes one or more electro mechanical devices, such as solenoids, that when engaged drive cartridges **90** outward until they engage the soil. Any other arrangement or

configuration or system is hereby contemplated for use as biasing member **172**.

(224) Alternative Graphene and/or Carbon Layers:

(225) In the arrangement described herein, graphene layer **116** is formed by applying a laser to the polyimide layer **114**. However, any other manner, method or means of forming and/or connecting graphene layer **116** to the underlying reference sensor **108** or nutrient sensor **110** is hereby contemplated for use.

(226) As one example, as an alternative, graphene flakes are overlaid on the exterior facing surface of reference sensor **108** or nutrient sensor **110** which are treated with an adhesive, which in one arrangement is an acrylic adhesive. Then pressure is applied to ensure a tight fit and secure attachment between the graphene flakes of the graphene layer **116** and the underlying reference sensor **108** or nutrient sensor **110**. This may provide a more-conductive graphene layer **116** and a more-stable manufacturing process.

(227) As another example, as another alternative, graphene flakes are overlaid on the exterior facing surface of reference sensor **108** or nutrient sensor **110**. Then a laser, which in one arrangement is a CO2 laser is applied over some or all of the area covered by the graphene flakes. In doing so, a portion of the material of the reference sensor **108** or nutrient sensor **110** are melted and/or mixed, and/or fused and/or welded together with the graphene flakes of graphene layer **116** thereby securely affixing the graphene layer **116** to the underlying reference sensor **108** or nutrient sensor **110**. This may provide a more-conductive graphene layer **116** and a more-stable manufacturing process.

(228) These manufacturing processes may be less prone to oxidation. Since reference sensor **108** and nutrient sensor **110** operate at very low voltage, in the range of millivolts and microvolts, any oxidation may cause reading inaccuracies and/or reading drift over time.

(229) As another alternative manner of attaching graphene layer **116** to the underlying reference sensor **108** or nutrient sensor **110**, electrically conductive transfer tape is used to transfer the graphene layer **116**, which may be formed of a LIG sensor formed from Polyimide tape onto the underlying gold pads of reference sensor **108** or nutrient sensor **110**. In this situation the sensor themselves would not have any polyimide components as the graphene layer **116**, which is LIG, is transferred from a polyimide tape to the underlying gold pads of reference sensor **108** or nutrient sensor **110**.

(230) To be clear, graphene and polyimide and Kapton are carbon based layers. As such, what is described herein is a carbon based conductive sensor that is configured to detect the presence of ions through changes in the electrical potential of the sensor. These changes in the electrical potential can be translated in to concentration of the particular sensed nutrient. As such, any form of a carbon based conductive sensor is hereby contemplated for use with the use of graphene, polyimide, Kapton, etc., simply being examples of carbon layers that may be used. That is, in this case, reference sensor or nutrient sensor **110** includes a carbon based layer that facilitates a carbon based conductive sensor wherein the electrical potential changes when ions engage carbon based conductive sensor. Additional forms of a carbon layer that may be used may include biochar, carbon black or any other form of a carbon layer.

(231) Testing Water Quality:

(232) Although some various disclosed embodiments may be primarily described in the context of soil measurements, the embodiments are not so limited. Rather, it is appreciated that the embodiments may be adapted for use in various other sensor applications. As one example, in one or more embodiments, soil moisture and fertility sensor system **10** may be adapted for use for sensing of quality and contents of water. For instance, in one or more arrangements, system **10** may be adapted for testing nutrient levels (e.g., nitrates) in various water sources including but not limited to, for example, tile drainage, irrigation (surface, canal, and/or well), rivers and streams, municipal water, and/or any other water source.

(233) Objectives Met:

(234) From the above discussion it will be appreciated that the soil moisture and nutrient sensor system presented herein improves upon the state of the art.

(235) Specifically, the soil moisture and nutrient sensor system presented in one or more arrangements: is more environmentally friendly than existing systems and methods; is more efficient than existing systems and methods; is safer to use than existing systems and methods; provides comprehensive moisture data across a field or farm; provides comprehensive moisture data at varying depths; provides comprehensive nutrient data across a field or farm; provides comprehensive nutrient data at varying depths; provides real-time data; provides actionable data; provides leading-indicator data; allows a farmer to be proactive instead of reactive; allows a farmer to maximize output; allows a farmer to minimize inputs; allows a farmer to optimize inputs and outputs; gives a farmer unprecedented visibility to soil moisture and nutrients; improves yields; reduces costs; is relatively inexpensive; is simple to use; can be used to sense nutrients for an entire growing season; has replaceable nutrient sensors; is wireless; is battery powered; helps a farmer meet increasingly strict environmental regulations; is easy to use; is easy to implement; has a robust design; provides charts and graphs of soil moisture and soil nutrients; provides a farmer with increased confidence in their decisions; provides accurate soil moisture readings; provides accurate soil nutrient readings; allows for the measurement of multiple nutrients; normalizes moisture and nutrient readings with temperature; provides co-located moisture, temperature and nutrient readings at various depths; is cost effective to use; can be used with any crop; can be used with any nutrient; provides highly repeatable results; has a robust design; provides highly accurate results; has a long useful life; is easy to install; has relatively few components; has a minimum number of parts; minimizes nutrient runoff; is environmentally friendly; improves a farmer's efficiency; saves a farmer time; and/or provides scientific information regarding soil moisture and nutrients, among countless other advantages and improvements.

(236) It will be appreciated by those skilled in the art that other various modifications could be made to the device without parting from the spirit and scope of this disclosure. All such modifications and changes fall within the scope of the claims and are intended to be covered thereby.

Claims

1. A soil moisture and fertility sensor system, comprising: a probe; the probe extending a length from an upper end to a lower end; the probe having a plurality of sensor modules; wherein the plurality of sensor modules are spaced along the length of the probe; wherein the plurality of sensor modules include a co-located moisture sensor assembly, a temperature sensor assembly and a fertility sensor assembly; wherein the probe is configured to be inserted into soil; wherein when the probe is inserted into soil, the probe makes a moisture measurement, a temperature measurement, and a fertility measurement of the soil at varying depths; wherein the fertility sensor assembly of the plurality of sensor modules include at least one nutrient sensor, wherein the at least one nutrient sensor includes a barrier having a selectively permeable membrane.
2. The system of claim 1, wherein the selectively permeable membrane is configured to permit a selected set of materials, ions, nutrients, chemistries, molecules, or constituents to be measured to pass through the selectively permeable membrane while blocking other ions, nutrients, chemistries, molecules, or constituents.
3. The system of claim 1, wherein the barrier includes a layer of material having one or more perforations.
4. The system of claim 1, wherein the barrier includes a layer of material having one or more perforations; wherein the one or more perforations are micro perforations.
5. A soil moisture and fertility sensor system, comprising: a probe; the probe extending a length from an upper end to a lower end; the probe having a plurality of sensor modules; wherein the

plurality of sensor modules are spaced along the length of the probe; wherein the plurality of sensor modules include a co-located moisture sensor assembly, a temperature sensor assembly and a fertility sensor assembly; wherein the probe is configured to be inserted into soil; wherein when the probe is inserted into soil, the probe makes a moisture measurement, a temperature measurement, and a fertility measurement of the soil at varying depths; a processing circuit; wherein the processing circuit is configured to normalize the fertility measurement of one of the plurality of sensor modules based on the temperature measurement of the one of the plurality of sensor modules to determine a normalized fertility measurement; wherein the processing circuit is configured to normalize the moisture measurement based on the temperature measurement to determine a normalized moisture measurement of the one of the plurality of sensor modules.

6. The system of claim 5, wherein the moisture sensor assembly of the plurality of sensor modules is formed of a capacitance sensor, wherein the capacitance sensor measures the dielectric constant of the soil.

7. A soil moisture and fertility sensor system, comprising: a probe; the probe extending a length from an upper end to a lower end; the probe having a plurality of sensor modules; wherein the plurality of sensor modules are spaced along the length of the probe; wherein the probe is configured to be inserted into soil; wherein when the probe is inserted into soil, the probe makes a moisture measurement and a fertility measurement of the soil at varying depths; wherein at least one of the plurality of sensor modules includes a moisture sensor assembly and a fertility sensor assembly; the moisture sensor assembly formed of a capacitance sensor having a positive plate, a negative plate and a space between the positive plate and the negative plate, wherein the fertility sensor assembly is positioned within the space between the positive plate and the negative plate.

8. The system of claim 7, wherein the fertility sensor is positioned within the moisture sensor assembly.

9. The system of claim 7, wherein the fertility sensor assembly includes a reference sensor.

10. The system of claim 7, wherein the fertility sensor assembly includes at least one nutrient sensor.

11. The system of claim 7, wherein the fertility sensor assembly includes at least one nutrient sensor, wherein the at least one nutrient sensor includes a polyimide layer.

12. The system of claim 7, wherein the fertility sensor includes at least one nutrient sensor, wherein the at least one nutrient sensor includes a layer of material having at least one perforation.

13. The system of claim 7, wherein the fertility sensor assembly includes at least one nutrient sensor, wherein the at least one nutrient sensor includes a graphene layer.

14. The system of claim 7, wherein the fertility sensor assembly includes at least one nutrient sensor, wherein the at least one nutrient sensor includes a layer of ceramic or glass that prevents soil adhesion.

15. The system of claim 7, wherein the fertility sensor assembly of the plurality of sensor modules include a cartridge that fits within a receptacle in the probe.

16. A soil moisture and fertility sensor system, comprising: a plurality of probes; a gateway; a database; the gateway communicatively connected to each of the plurality of probes and the database; wherein the gateway is configured to receive soil measurement data from each of the plurality of probes; wherein each of the sensor modules is configured to measure soil measurement data at different depths and communicate the soil measurement data to the gateway; wherein the gateway is configured to communicate the soil measurement for the different depths received from the plurality of probes to the database.

17. The system of claim 16, further comprising: a processing circuit communicatively connected to the database; wherein the process circuit is configured and arranged to evaluate the soil measurement data in the database and identify trends in the soil measurement data.

18. The system of claim 16, further comprising: a processing circuit communicatively connected to the database; wherein the process circuit is configured and arranged to evaluate the soil

measurement data in the database and predict future soil conditions that will result from a proposed soil treatment specified by a user.

19. The system of claim 16, further comprising at least one additional sensor module; wherein the at least one additional sensor module is configured to measure soil measurement data from a single soil depth and communicate the soil measurement data to the gateway.

20. A sensor system, comprising: a probe; the probe extending a length from an upper end to a lower end; the probe having a plurality of sensor modules; wherein the plurality of sensor modules are spaced along the length of the probe; wherein the plurality of sensor modules include a fertility sensor assembly; wherein the probe makes a fertility measurement at varying points along the length of the probe; wherein the fertility sensor assembly of the plurality of sensor modules includes at least one nutrient sensor, wherein the at least one nutrient sensor includes a layer of material having at least one perforation.

21. The system of claim 20, wherein the probe is configured for taking measurements in soil.

22. The system of claim 20, wherein the probe is configured for taking measurements in a body of water.
